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الغذاء
LAEQ
FOOD
SAFETY

Level 3 in Food Safety

Supervision of Food Safety



Course Outline

01 Introduction to Food Safety.

Hazards Contamination and control 02

03 Microorganisms (germs)

04 Food poisoning and foodborne illness

05 Personal hygiene and health requirements

06 Risks and control from receiving to serving of food

07 Food spoilage and food preservation

08 Design and use of food premises and equipment

09 Cleaning and disinfection

10 Pest control

11 HACCP and Food Safety Management Systems.

12 Quality and Food Safety Policy

13 Food safety legislation and standards

Course Outcomes:

Delegates will be able to:

- Improve the standards and safety of their operation
- Establish and maintain prerequisite programs
- Comply with food safety legislation
- Apply the basic principles of HACCP
- Confidently deal with enforcement officers
-Successfully complete a level 3 Award in Supervising Food Safety examination

Goals

The course aims to qualify supervisors of food establishments and managers to achieve effective supervision of food and abide by the requirements.



Module 1



Introduction to Food Safety

Definitions

Carriers

Is a person who does not show symptoms of poisoning but carries the causative microbe responsible for the illness and can be transmitted to others.

Contamination

Is the presence of the dangerous substance in the food causing poisoning or illness.

Cross contamination

Is the transmission of germs from raw food to Ready-to-eat food either directly or indirectly

Food Safety

Are the necessary measures and conditions that are established to make sure that food is safe for human consumption.

Food Poisoning

It is a medical condition that occurs when eating food that is contaminated by food-borne pathogens or toxins

Definitions

Food safety hazards

The possibility of posing a danger to food safety whether biological, physical, chemical or allergenic.

Food Safety Management System

Set of policies, processes, and documents that ensure that the food that is sold and consumed is free of hazards

High-risk foods

Ready-to-consume cooked food that allow the growth of pathogenic germs under favorable conditions

Pathogen

It is a microorganism that causes diseases in humans

Safe Food

It is food free of contaminants that will not causes health damage to the consumer.

What do you think are the most common hygiene faults that result in food poisoning?

-Preparation too far in advance and storage at room temperature .
- Slow cooling .
- Inadequate reheating/cooking .
- Contaminated food (cross-contamination or raw).
- Inadequate thawing prior to cooking .
- Food handlers (infected/bad personal hygiene).

What are the benefits of good hygiene?

- A good reputation
–customer confidence
- Improved food safety standards
- Compliance with food safety legislation
- Reduced risk of food poisoning
- Longer shelf-life
-Good working conditions, higher staff morale and reduced turnover of staff
- Increased productivity
- Brand protection
- Increased business
–higher profits

What are the costs of poor hygiene?

- A bad reputation
- Food contamination and customer complaints
- Fines and legal costs resulting from prosecution
- Civil action from ill people and/or customers
- Loss of business, paid by the consumer
- Increased risk of food poisoning
- Death and suffering
- Increased risk of pest infestation
- Waste food due to spoilage
- Poor morale-high staff turnover
- Loss of business –lower profits



Module 2

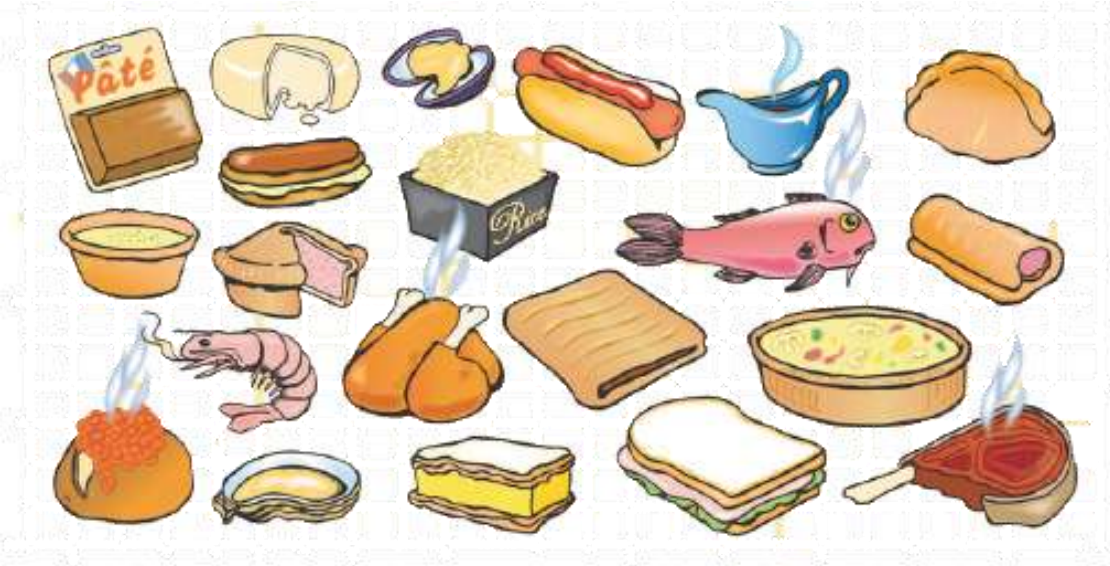


Hazard Contamination and Control

High-risk Foods

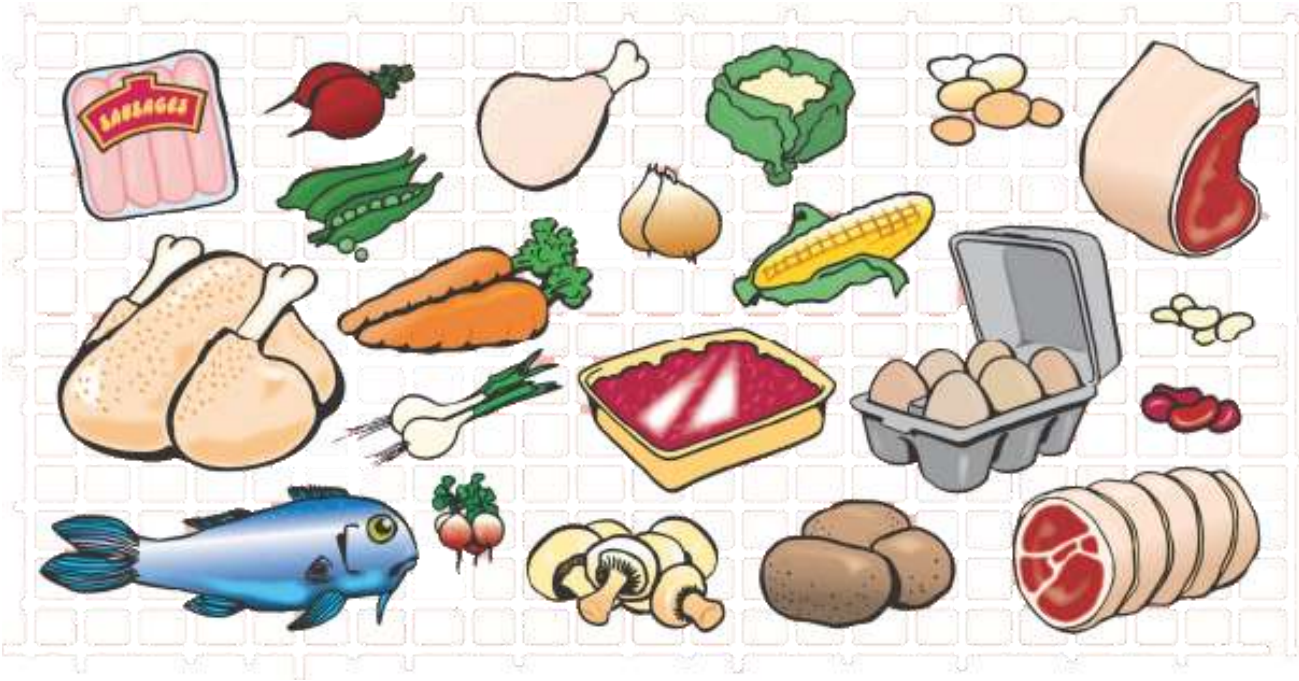
-High-risk foods are ready-to-eat foods, which support the multiplication of harmful bacteria and include most cooked foods.

- These foods are usually proteins.



Raw Foods to be Cooked

Raw foods to be cooked must be separated from ready-to-eat food. They are a major source of food poisoning bacteria



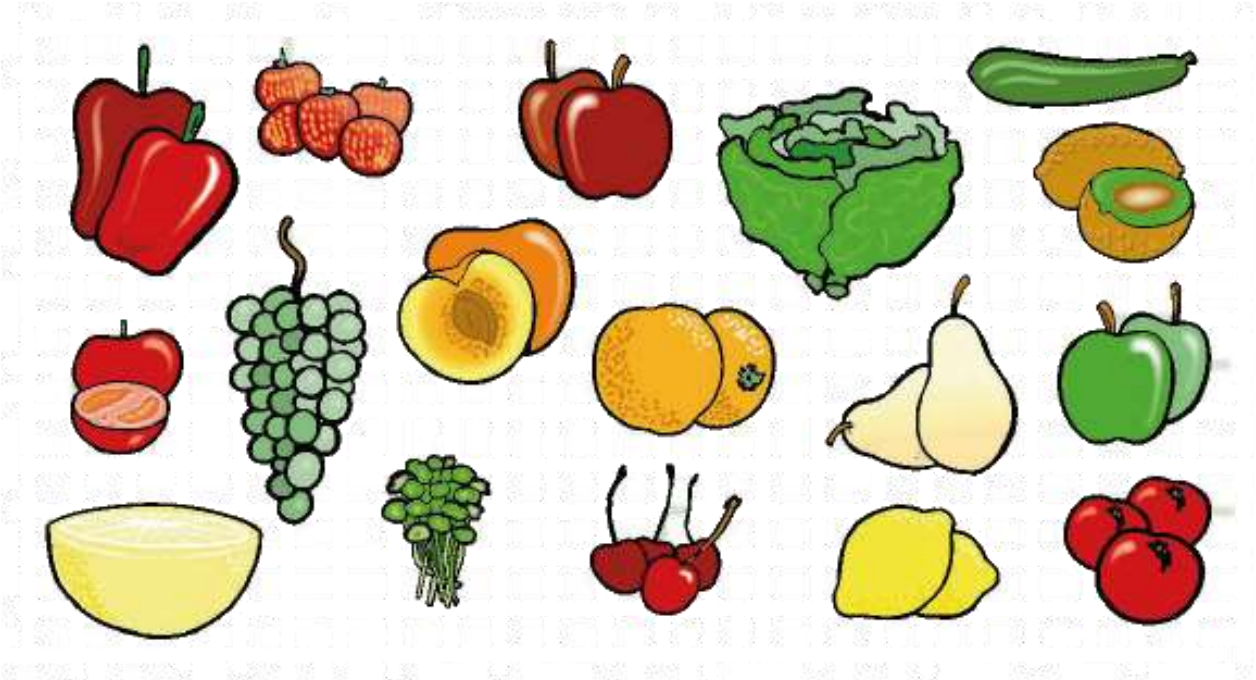
Low-risk Foods

Low-risk food can usually be stored at ambient/room temperatures as they do not support the multiplication of pathogenic bacteria



Ready-to-eat Raw Foods

Should always be washed and if possible scrubbed before eating as some low dose bacteria may be present



Choose and distinguish high-risk foods with a correct sign

Cream or custard ✓



Flour



Arugula salad ✓



Orange



Raw egg products ✓



Raw chicken



Lettuce ✓



Dried fruits



Smoked chicken breasts ready for consumption ✓

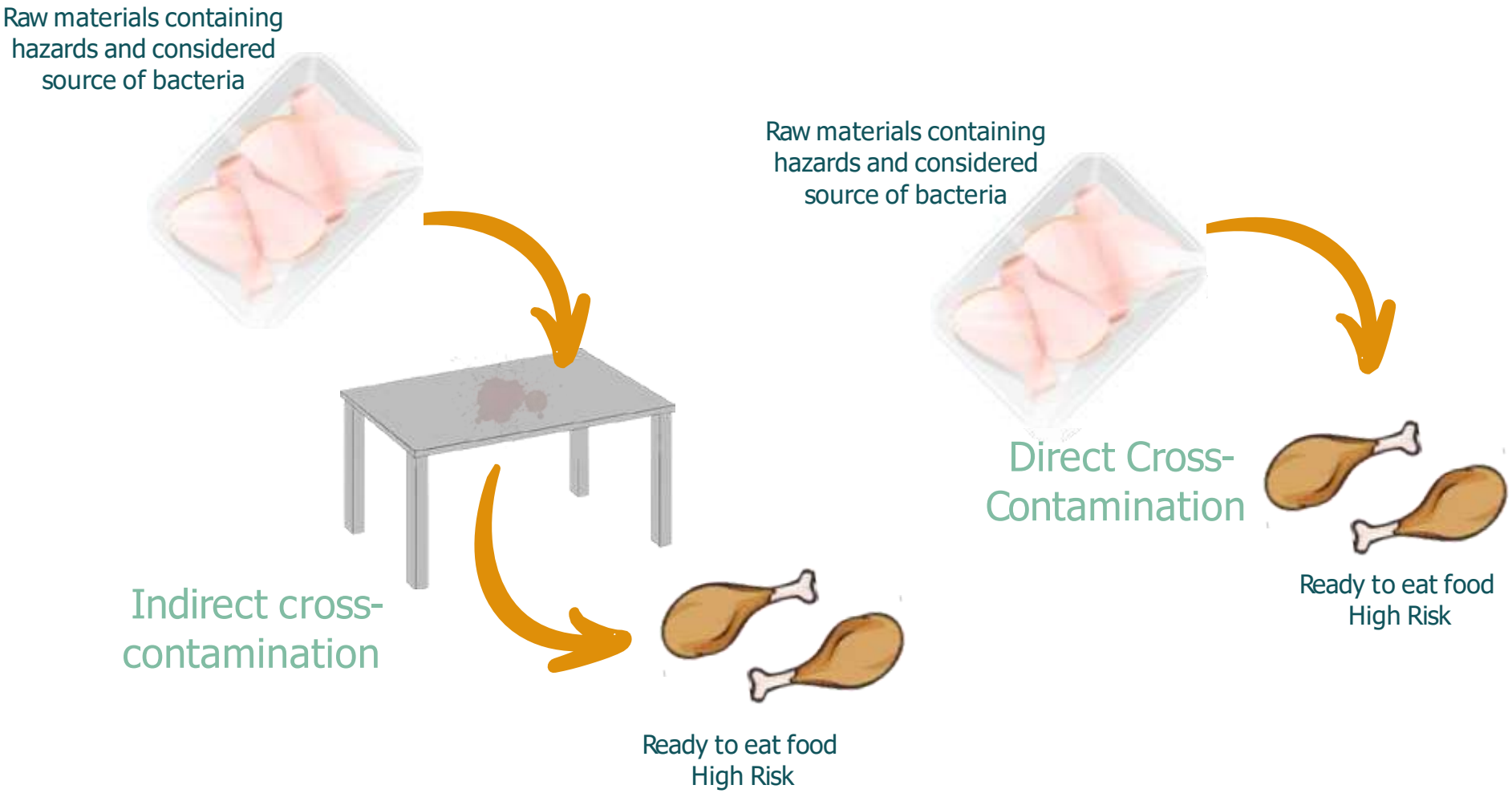


Types of contamination

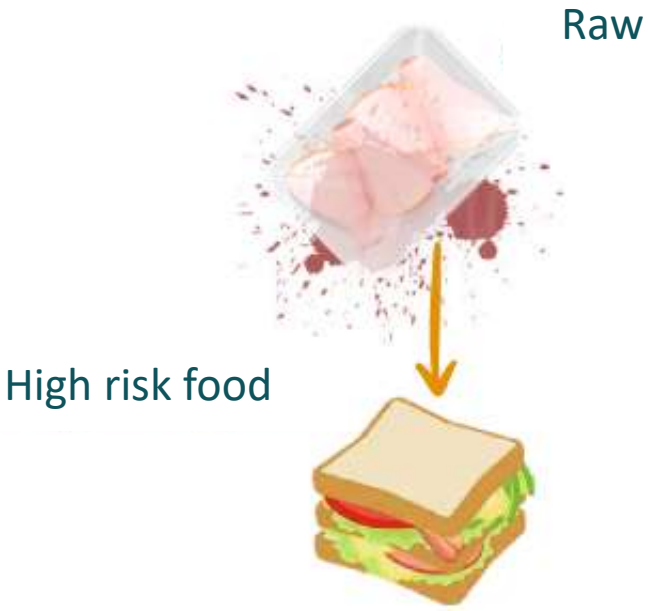
Contamination: the presence or introduction of any hazardous substance to food

Microbiological hazards	Physical hazards	Chemical hazards	Allergenic hazards
Bacteria(pathogenic bacteria or Spoilage bacteria) viruses, .spores, fungi or mold	Glass, Hair, Plastic	Excessive preservatives ,than recommended Residue of detergents, pesticide residues	Allergic ,eggs, milk, fish nuts, gluten

Cross - Contamination



Direct Cross-contamination



Thawing liquid

What are the preventive measures from microbiological contamination?

Elimination of the source of hazard	Cut any routes leading to the contamination from the hazard	Elimination of contaminants of food
-By purchasing from an approved supplier to avoid raw materials contamination - Check the safety of materials upon receiving - Check food transportation trucks against food safety requirements (temperature, hygiene) - Immediate storage. -Put in place a pest control program in the facility. - Maintain personal hygiene. - Develop policies for visitors. - Staff training.	- Instructions, supervision and effective training. - Good design of buildings and equipment. - Good hygiene practices. -Protect food with storage ,packaging and reduce handling. -Separation of high-risk food (ready to eat foods) and raw food (color coding). - Efficient cooling and defrosting systems. -Eliminate contaminated/expired foods immediately or separate them from safe foods. - Develop an effective cleaning schedule.	-Effective cooking with sufficient temperatures and time. - Disinfection. - Food processing. - Proper disposal of contaminated food.

What are preventive measures from physical contamination?

Physical hazards – when food (raw or ready to eat) is exposed to the following:

- Hair
- Wood
- Insects and rodents parts
- Plastic ,etc

Protection is by:

- Choosing an approved supplier.
-Inspection of raw materials as soon as they are received.
-Wear protective clothing to avoid contamination of hair or nails while preparing, transporting and storing raw materials.



What are preventive measures from physical contamination?

Physical hazards from packaging by: materials:

Pins
Cardboard
Threads and cloth cotton
Glass
Rubber bands
Plastic
Wood
Polyethylene Paper

Protection is

- Caution during unpacking.
-Caution while disposing of packaging materials.



What are preventive measures from physical contamination?

Physical hazards from equipment & utensils:

- Wood
Screw drives and screws Stones
Glass
Paint chips
Rust
Grease / oil (used in lubrication and maintenance of equipment)

Protection is by:

- Caution during maintenanc operations and performing them outside of working hours.
- Avoid temporary repairs.
-Use tools of high-quality materials that is suitable for use for foods
- Avoid using wood.
- Develop policies for handling glass



What are preventive measures from physical contamination?

Physical hazards from employees & visitors:

Hair
Nails
Jewelry
Pieces
Buttons
Pen covers
Cigarette butts



Protection is by:

- Good training of workers.
-High standards of personal hygiene.
- Wear protective clothing.
- Develop policies for visitors.

What are preventive measures from physical contamination?

Physical hazards from pests:

Parts of rodents and insects Waste
Hair
Eggs
Larvae
Feathers

Protection is by:

Contract with an approved pest control company.



What are the sources of chemical hazards?

Chemical hazards in raw materials:	Chemical hazards during food handling and processing:
Pesticides / Herbicides / Fungicides / Manure (fruits / vegetables / cereals) Minerals (fish, citrus fruit) Antibiotics/Hormones (Meat) Natural toxins (red beans) Industrial Chemicals	Vapor Cleaning Chemicals Excess additives Chemicals from packaging materials

What are preventive measures of chemical hazards?

- Approved suppliers of cleaning materials and raw materials.
- Safe packaging materials.
- Separation of chemicals from foods.
- Avoid storing food, fat, and beverages in old chemical cans.
-Avoid storing chemicals in cans whose contents are not shown and that are not labeled
- Correct dilution of cleaning agents.
- Proper rinsing after cleaning with chemicals.
- Food protection/covering
-Provide approved detergents that are safe to use in food facilities –food grade chemicals

Allergenic Hazards

Food allergy: It is when the immune system gives an unexpected reaction to certain nutrients

Allergy symptoms:

Although some of the symptoms are mild, some allergic reactions can be very dangerous.

A local study conducted by the National Committee for Nutrition at the Food Authority found that from a sample size of 15142 21% suffered from food allergies.



Symptoms of allergenic reactions:

- Tightness and itching in the throat
- Redness and itching of the skin
- Difficulty swallowing
- Difficulty breathing
- Nausea and vomiting
- Abdominal pain
- Diarrhea
- Sneezing and itchy eye

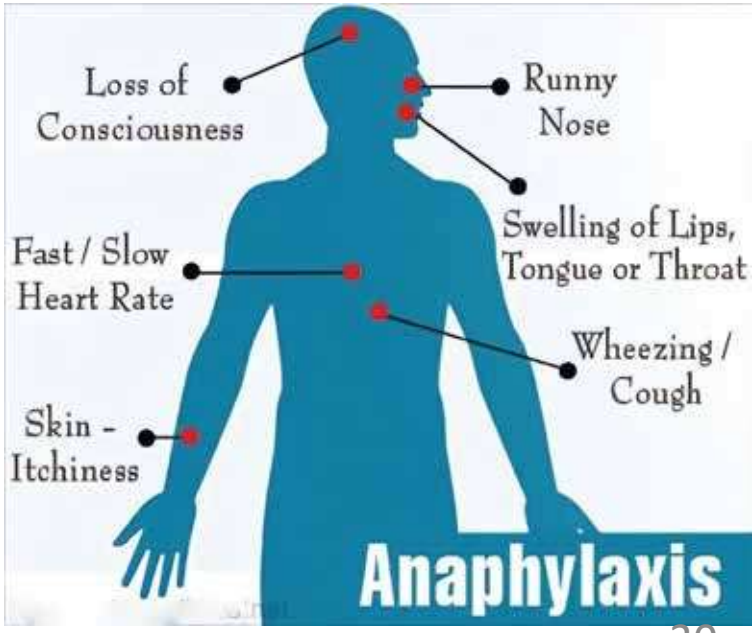


Symptoms of allergenic reactions:

Anaphylaxis: It is a life-threatening allergenic reaction that puts the body in a shock.

At the beginning, the symptoms of anaphylaxis are/ can be:

Difficulty breathing
Frequent cough
Weakness in pulse
Skin allergy
Tightness of the chest
And symptoms in several parts of the body



Food Intolerance

Food intolerance: A food intolerance is difficulty digesting certain foods and having an unpleasant physical reaction to them. It causes symptoms, such as bloating and tummy pain, which usually happen a few hours after eating the food



Facts About Food Allergy

- Even the tiniest amount of a food allergen can cause a reaction.

-Some people are more susceptible to being allergic to certain foods.

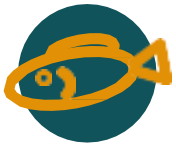
- There is a difference between food allergy and food intolerance.

-Food allergy is very serious and controls must be put in place throughout food production from farm to fork.

The most common types of food allergens :

in the Kingdom as per a national study carried out by the Saudi Food and Drug Authority:


Cereals and gluten


Fish


Peanut


Tree Nuts


Lupin


Milk


Sulfites and Sulphur dioxide


Celery


Crustaceans


Soybeans and Products


Eggs


Mustard


Sesame Seeds


Molluscs

What are preventive measures from allergenic hazards?

- **Separation of allergens from non-allergenic food items**
- **Use separate tools to prepare food free of allergens**
- **Color coded tools that distinguish the purpose of their use**
- **Labels on the products prominently showing their ingredients**
- **Effective cleaning**

Duties of employees

- Hand washing before preparation
- Commitment to the separation of allergenic tools
- Awareness of allergy symptoms and reporting to the supervisor as soon as they occur

Module 3



Microorganisms (Germs)

Microorganisms (germs)

- Living organisms
-Not visible to the naked eye (only under a microscope), micro=small.
-Called microorganisms, including: bacteria, viruses and fungi.

Bacteria

They are found everywhere.
- Disease causing (pathogens)
- Beneficial
- Essential
- Spoilage



Requirements for the multiplication of bacteria

Temperature: Low temperatures are 1-4°C (refrigerator or chiller) prevents the multiplication of some types of bacteria. Yet, spoilage bacteria can multiply inside the refrigerator and hence spoil food.

Time: Under favorable conditions, bacteria can divide every 10 minutes.

Moisture: Bacteria require water to transport nutrients into the cell and take away waste products. Bacteria also moves in water by whips so the reduction in water ctivity (by drying) prevents its movement.

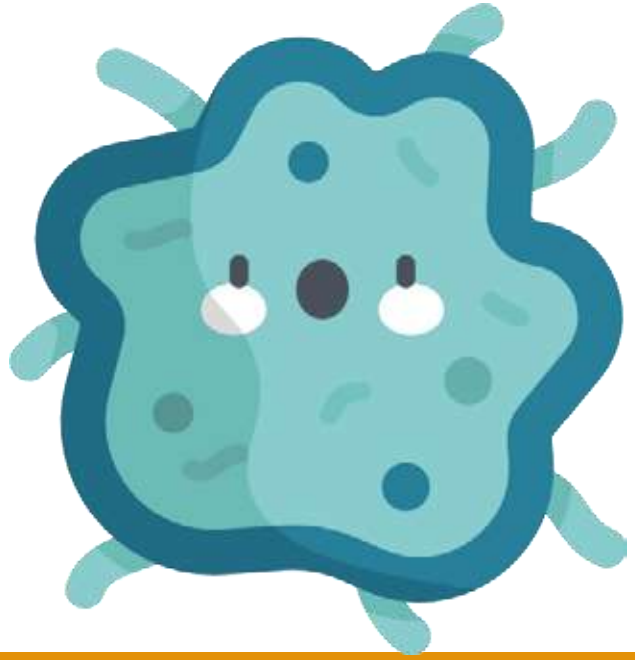
Food



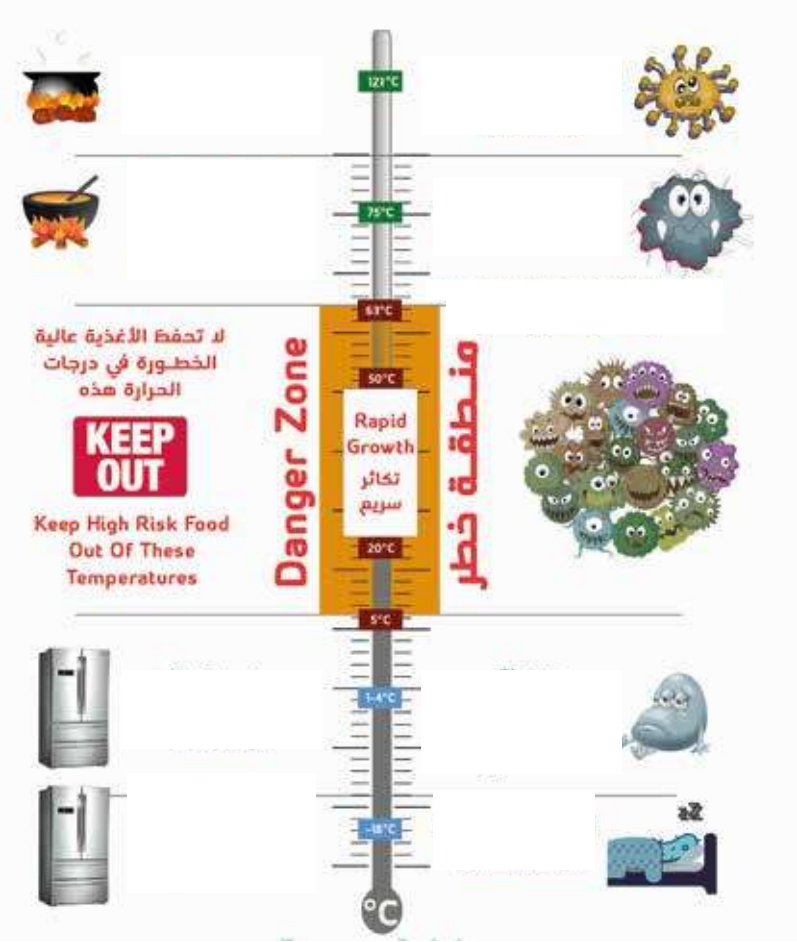
Requirements for the multiplication of bacteria

Types of bacteria depending on the temperatures at which they survive:

Thermophiles survive at elevated between 45--120℃ temperatures (Thermophile)
Mesophiles survive between 20--45 ℃
Psychrophiles survive at temperatures below 20 ℃.



Temperature and growth of microorganisms



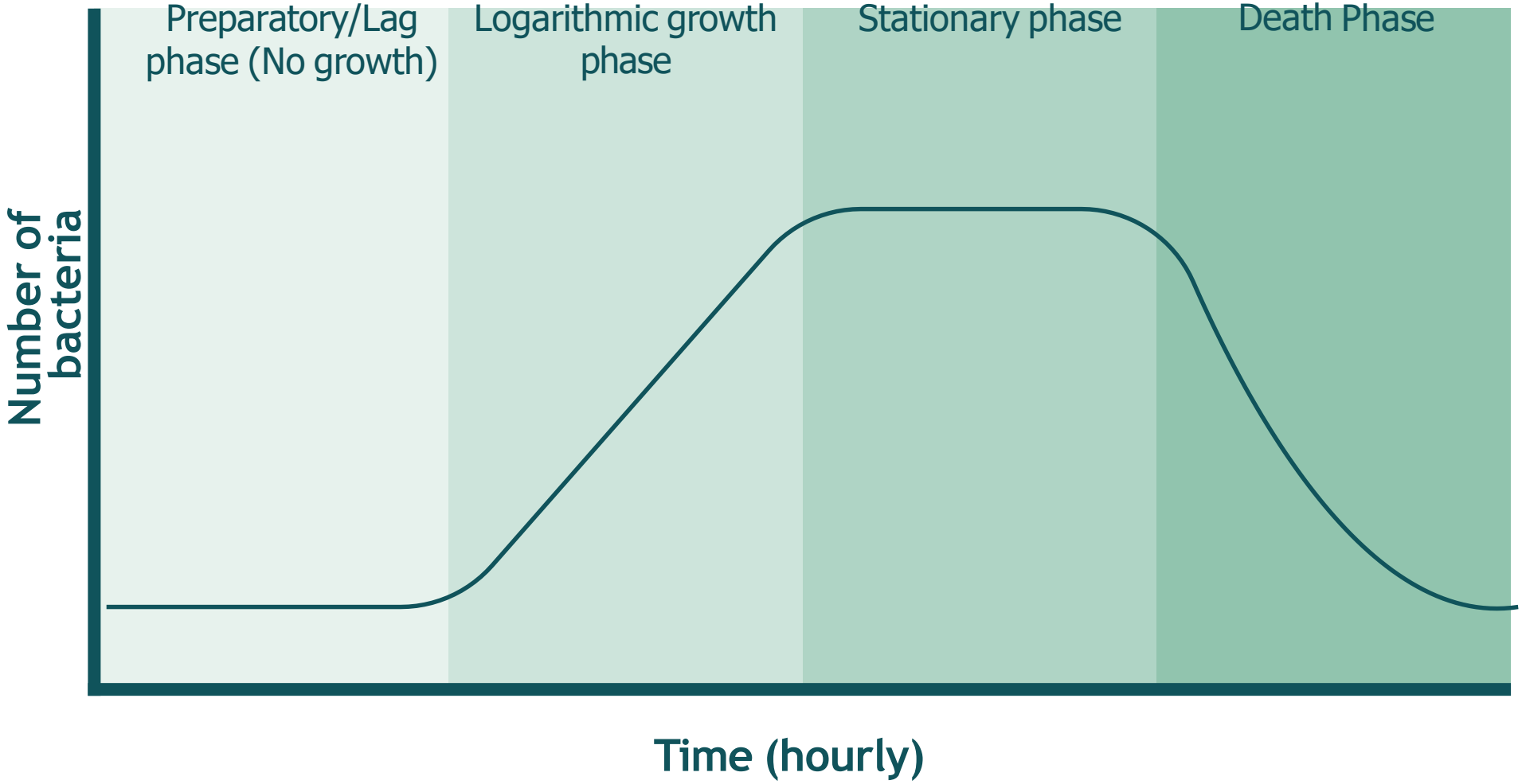
At temperatures above 63 °C most microbes die

Danger Zone temperature 5 - 63 °C

C reduce growth to the °5-Temperatures between 0 greatest degree, Possibility of growth of some food-spoiling bacteria without harm

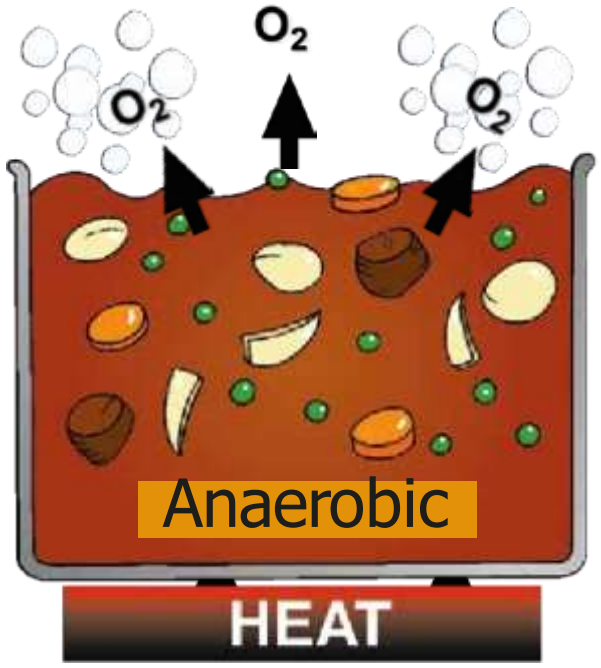
Temperatures below 0 °C inhibit the growth of bacteria

Bacterial growth curve



Oxygen and bacteria

- Most types of bacteria do not need oxygen
- Aerobic bacteria need oxygen to grow
- Anaerobic bacteria grow only when there is no oxygen



Elimination of bacteria in food

HTST: (High Temperature Short Time) HTST pasteurization process is a continuous process accomplished by heating the product to certain temperature and passing through a holding tube at a rate that ensures the required holding time.

LTLT: Low-Temperature Long-Time pasteurization is a pasteurization technique commonly used in small-scale and artisanal operations. This process is also called “batch” pasteurization or “vat” pasteurization.

UHT: Ultra-high temperature pasteurization. Ultra-high-temperature (UHT) pasteurization involves heating milk or cream at high temperatures for one or two seconds. Packaged in sterile, hermetically sealed containers, UHT milk may be stored without refrigeration for months.

Canning: designed to destroy of Clostridium botulinum spores.

Effective cooking: the core temperatures of the food reaches 75 °C.

Chemical substances

Irradiation
Ultraviolet (UV) rays

Spores

-Hibernation phase resistant to unfavorable growth conditions

- Can survive:

Surviving at high temperatures, such as boiling for hours 5

Drying

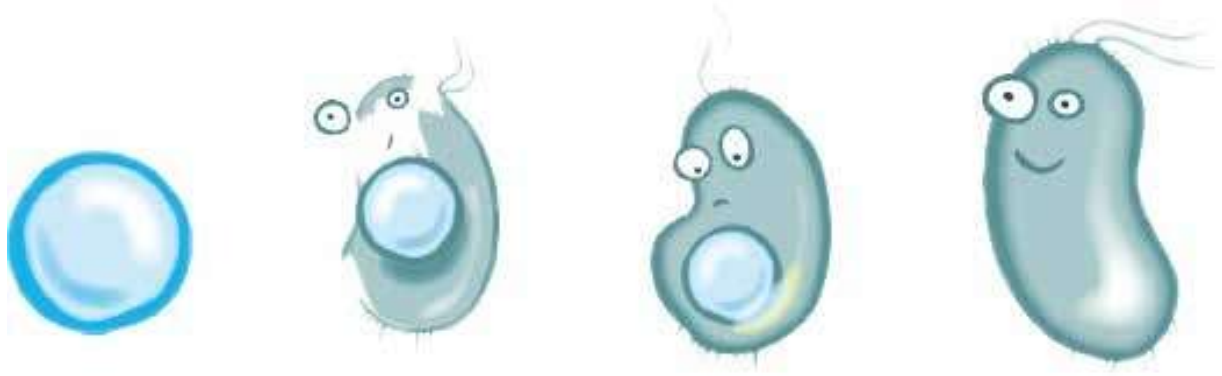
Lack of food

UV rays

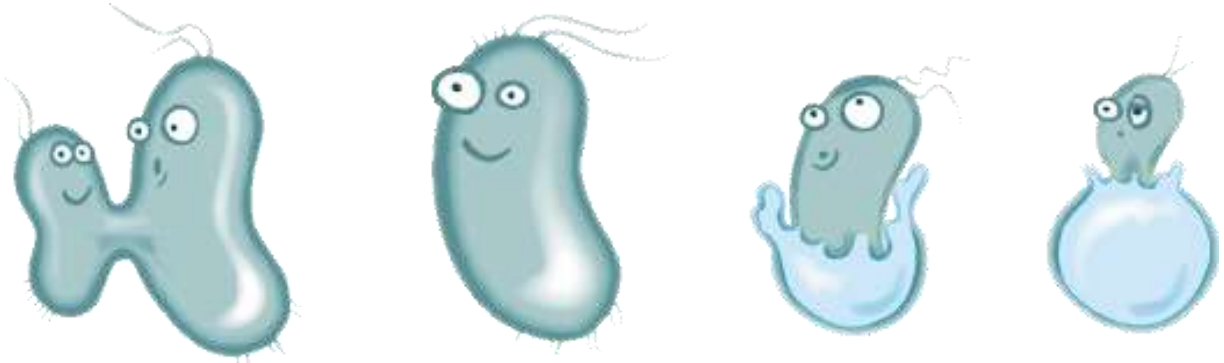
Disinfectant

Do not multiply





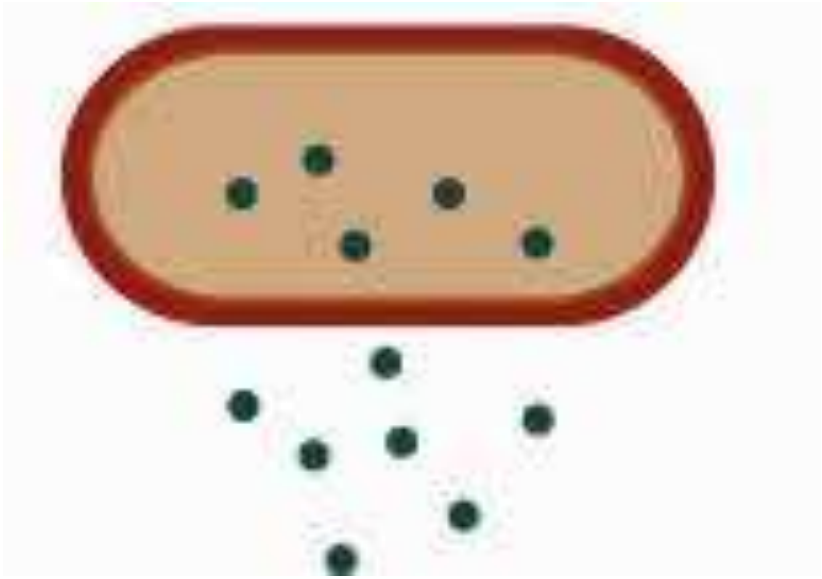
The spore forms inside the cell after which the cell decomposes and releases the microorganism and might secret toxins



When the favorable conditions are in place, the spore begins germination and the bacterial cell begins to multiply which can cause poisoning

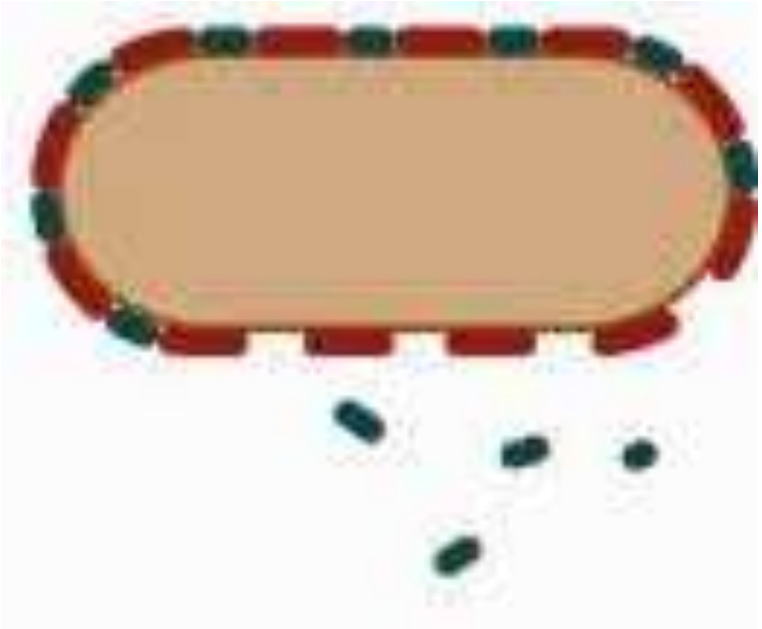
Exotoxin (toxins produced by bacteria)

A protein produced by bacteria inside the food is toxic.
Symptoms of food poisoning: Some types of toxins infect the digestive tract and can cause diarrhea and vomiting are caused by "Enterotoxin" and some types of toxins cause damage to the nervous system "Neurotoxins".



Endotoxin

Part of the membrane of bacteria that is released when bacteria die or are released by spores



Module 4



Food poisoning and foodborne illness caused by contaminated food

Definitions

Pathogens

Microorganisms that contaminate food and cause illness

contaminated food

Food that probably causes illness after being consumed

Incubation period

The period between consumption of contaminated food and the appearance of the first symptoms of illness.

Source

The point at which the pathogen enters the food chain or source through which it enters the food establishment.

Groups at risk

- Elder People
- Children and infants
- Pregnant women
-People with immunodeficiency or compromised immune system
- Ill people.



Pathogens in food

- Pathogenic bacteria
- Viruses
- Mold
- Protozoa



General properties of food poisoning

Incubation period

The incubation period varies depending on the bacterium causing the disease, but in general the period usually is between one hour and 36 hours.
- Some types of pathogenic bacteria have more extended incubation period:

.Salmonella: 6 hours - 6 days
.E.coli: Three to four days
.Listeria: Symptoms appear within two weeks

There are also other types whose incubation period is shorter:
- Staphillococcus aureus (Staphillococcus aures): 30 minutes to 8 hours.

Remember:

It is important to know the incubation periods of food pathogens to track food that caused the food poisoning and one must correct the error that led to the poisoning incident.

General properties of food poisoning

Duration of the disease

The duration of the disease varies from case .to case but usually is from a day to a week We usually need large numbers of food poisoning microbes (bacteria, viruses, etc.) to cause food poisoning. Not to mention, the entry of the pathogen and the multiplication inside the human body does not necessarily lead to food poisoning, on some cases it's not until damage to the cell happens for the immune system to react and lead to food .poisoning



Symptoms of food poisoning

Symptoms range from simple and acute, varies by the causative agent, may include:

- Abdominal pain
- Stomach cramps
- Nausea
- Vomiting
- Diarrhea
- Fever

When any of the symptoms occur, you should see a doctor:

- Bloody diarrhea.
- Temperature rise (38.8 degrees Celsius).
- Frequent vomiting prevents you from consuming fluids and may cause dehydration
- Signs of dehydration include decreased urination, mouth dryness, and feeling dizzy while standing.
- Diarrhea that lasts for more than 3 days.

Sources and symptoms of some of the most important foodborne pathogens that cause illness

Staphylococcus aureus

Incubation period:
1- 7 hours after ingestion

The most common sources:
Skin. Reheated foods, high in protein, meat, poultry, eggs, milk products, stuffings, prepared salads…

Symptoms
Nausea, vomiting, pain, diarrhea, NO FEVER

Sources and symptoms of some of the most important foodborne pathogens that cause illness

Bacillus cereus

Incubation period:
30 min - 6 hours (emetic), 6- 15 hours (diarrheal); lasts up to 24 hours

The most common sources:
Rice products (fried rice), starchy foods (potatoes, pasta, cheese products), food mixtures like casseroles, sauces, puddings, soups, pastries (emetic), meats, milk, vegetables, fish (diarrheal).

Symptoms
Emetic (nausea, vomiting); diarrheal (watery diarrhea, cramps)

Sources and symptoms of some of the most important foodborne pathogens that cause illness

Salmonella spp

Incubation period:

Symptoms appear 6 hours - 6 days after infection.

The most common sources:

raw chicken, turkey, chicken meat, milk and unpasteurized juice, and uncooked vegetables and fruits.

Symptoms

Diarrhea, fever, stomach cramps, vomiting.



Sources and symptoms of some of the most important foodborne pathogens that cause illness

Escherichia coli O157

Incubation period:
Symptoms appear 3- 4 days after infection.

The most common sources:
Raw minced meat, milk and unpasteurized juice, raw vegetables such as lettuce, contaminated water.

Symptoms
Diarrhea (usually bloody), severe stomach cramps, vomiting.

5 - 10% of people diagnosed with E. coli experience life-threatening health problems.

Sources and symptoms of some of the most important foodborne pathogens that cause illness

Listeria monocytogenes

Incubation period:
Symptoms appear within two weeks of infection.

The most common sources:
Non-hard cheeses, unpasteurized milk, sausages

Symptoms
Pregnant women who have a listeria infection experience symptoms like fever from high temperatures, fatigue and muscle pain. Infection during pregnancy may also cause serious diseases or even death of newborns.
Other sufferers (usually adults) experience symptoms that include headache, loss of balance, neck stiffness, cramps, muscle pain and fever

Sources and symptoms of some of the most important foodborne pathogens that cause illness

Clostridium perfringens

Incubation period:

Symptoms appear within 6 – 24 hours of infection.

The most common sources:

Meat and poultry, dried or pre-cooked food.

Symptoms

Diarrhea, stomach cramps usually start suddenly and last less than 24 hours.

Vomiting and fever are uncommon in people with this infection

Sources and symptoms of some of the most important foodborne pathogens that cause illness

Clostridium botulinum

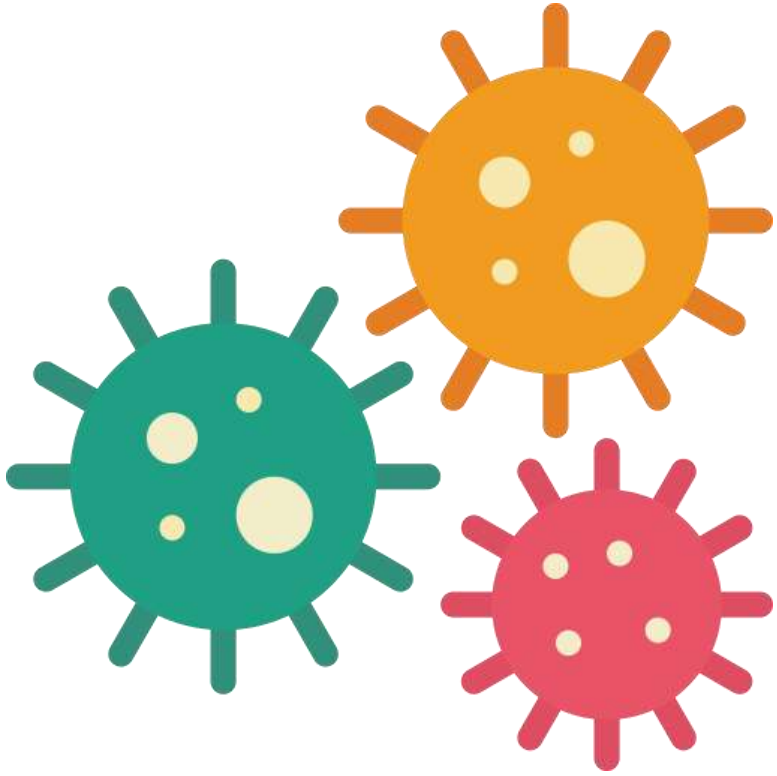
Incubation period:
Symptoms appear within 36-18 hours of infection.

The most common sources:
Honey, canned or wrongly brewed food (usually homemade)

Symptoms
Blurred vision, difficulty in speaking, swallowing and breathing, weakness and loss of control of muscles, symptoms begin with the head and then the rest of the body down, as the intensity increases.

Viruses

- Smaller than bacteria
- Multiply in living cells of body, not food
- Low infective dose
- Person to person spread common
- Environmental spread
- Destroyed by thorough cooking



Non-bacterial food poisoning

Poisonous chemicals

Fungicides, weedkillers, pesticides, cleaning, chemicals, additives, etc.

Poisonous metals

Antimony, cardmium, copper, iron, lead, mercury,tin, zinc, etc.

Poisonous plants

Deadly nightshade, death cap, daffodil bulbs, toadstools, rhubarb leaves, red kidney beans, etc.







Moulds (mycotocxin)

Nuts, e.g. peanuts. Cloudy apple juice.



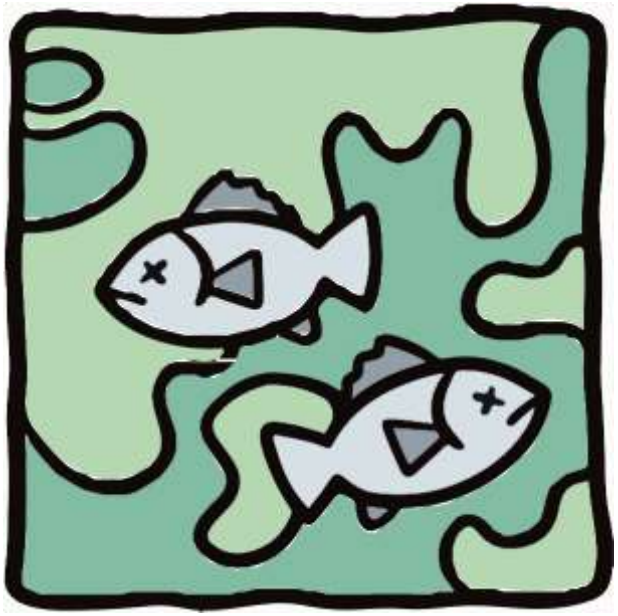
Dangerous Plants

- Bitter almonds
- Potatoes (green/sprouting)
- Deadly nightshade
- Red kidney beans
- Toadstools
- Rhubarb leaves
- Daffodil bulbs
- Death cap



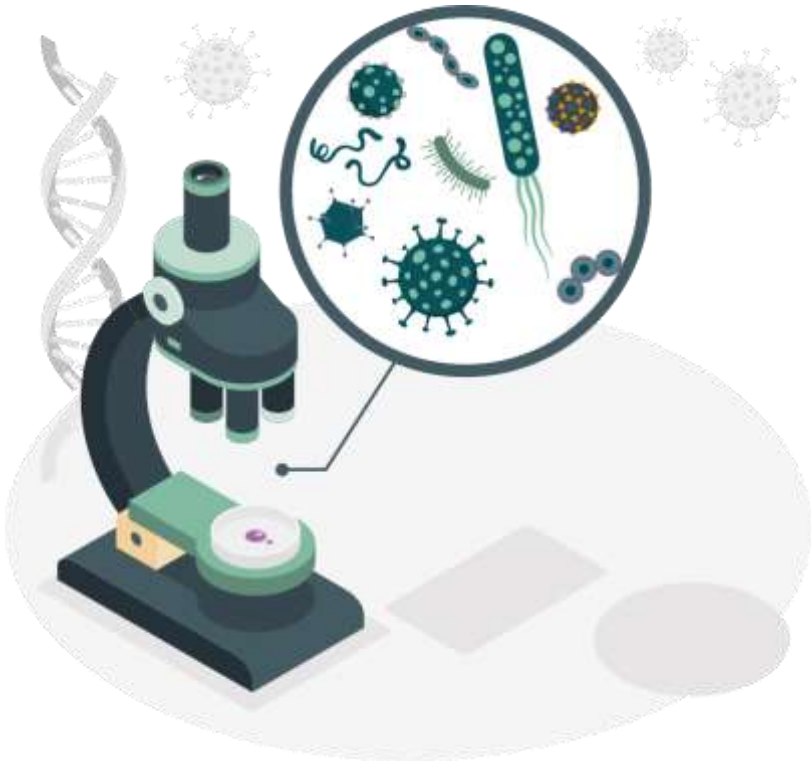
Fish/Shellfish Poisoning

- Scombrotoxin
- Ciguatera
- Diarrhetic shellfish poisoning (DSP)
- Fugu (puffer fish)
- Paralytic shellfish poisoning



Cause of foodborne illness

- **Campylobacter enteritis**
- **Escherichia coli O157**
- **Norovirus**
- **Listeria**
- **Typhoid/paratyphoid**
- **Hepatitis A**
- **Dysentary**
- **Parasites**
- **Low -dose organisms**
- **Small numbers**



- **Do not need to multiply in food**
- **Multiply in the body**

Prevention of food poisoning

-Breaking the food poisoning chain, because food poisoning is caused by a single mistake or a combination of mistakes
-Application of HACCP system (Hazard Analysis and Critical Control Point) or any food safety management system depending on the need of the facility.
- Removal of sources of hazards.
- Prevention of contamination of food.
- Prevent the growth/multiplication of bacteria.
- Elimination of bacteria.
- Training of employees and food handlers.

Food Poisoning Investigation

What is the role of the supervisor?

- Advise manager
- Suspend sales
- Remove and isolate
- Exclude staff
- Specimens from ill staff
- Foods implicated
- Complaints
- Food history
- Answer questions
- Provide records for audit
- Sampling
- Clean and disinfect
- Arrange for restock

Module 5



Personal hygiene and health requirements

Source of hazards related to food handlers

- Contaminated hands
- Illness, vomiting, diarrhea
- Wounds and scratches to the skin
- Open Wounds
- Negligence of personal hygiene
- Contaminated clothing
- Jewelry
- Nails
- Smoking



When should I wash my hands?

Before or after?





Applying bandages to wounds



Dealing with an ill person



Entering the food preparation areas



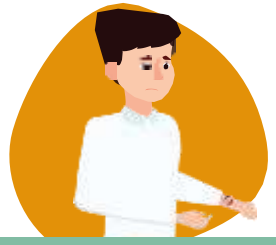
Touching dirty clothes or cartons

When should I wash my hands?

Before or after?



Always and continuously



After applying bandages to wounds



After dealing with an ill person



Before entering the food preparation areas



After touching dirty clothes or cartons

When should I wash my hands?



Dealing with ready-to-eat foods



Dealing with raw foods



Waste disposal



After using the toilets



Touching the face or hair or cleaning the nose

When should I wash my hands?



Before dealing with ready-to-eat foods



After dealing with raw foods



After waste disposal



After using the toilets



After touching the face or hair or cleaning the nose

Duties of food handlers

- To have high standards of personal hygiene.
- Commitment to wear clean protective clothing.
- Inform the supervisor in cases of illness - God forbid -



Wash your hands thoroughly



1

غسل اليد بالماء

Wash your hands with water



2

استخدم الصابون

Use soap



3

غسل كف الايدي

Rub palm to palm



4

دعك الكف بالكف

Back of hands



5

دعك الاصابع

Rub fingernails



6

تخليل الاصابع

Fingers interlaced

Wash your hands thoroughly



7

غسل الابهام

Base of thumbs



8

دعك المعصمين

Rub wrists



9

شطف الايدي

Rinse hands

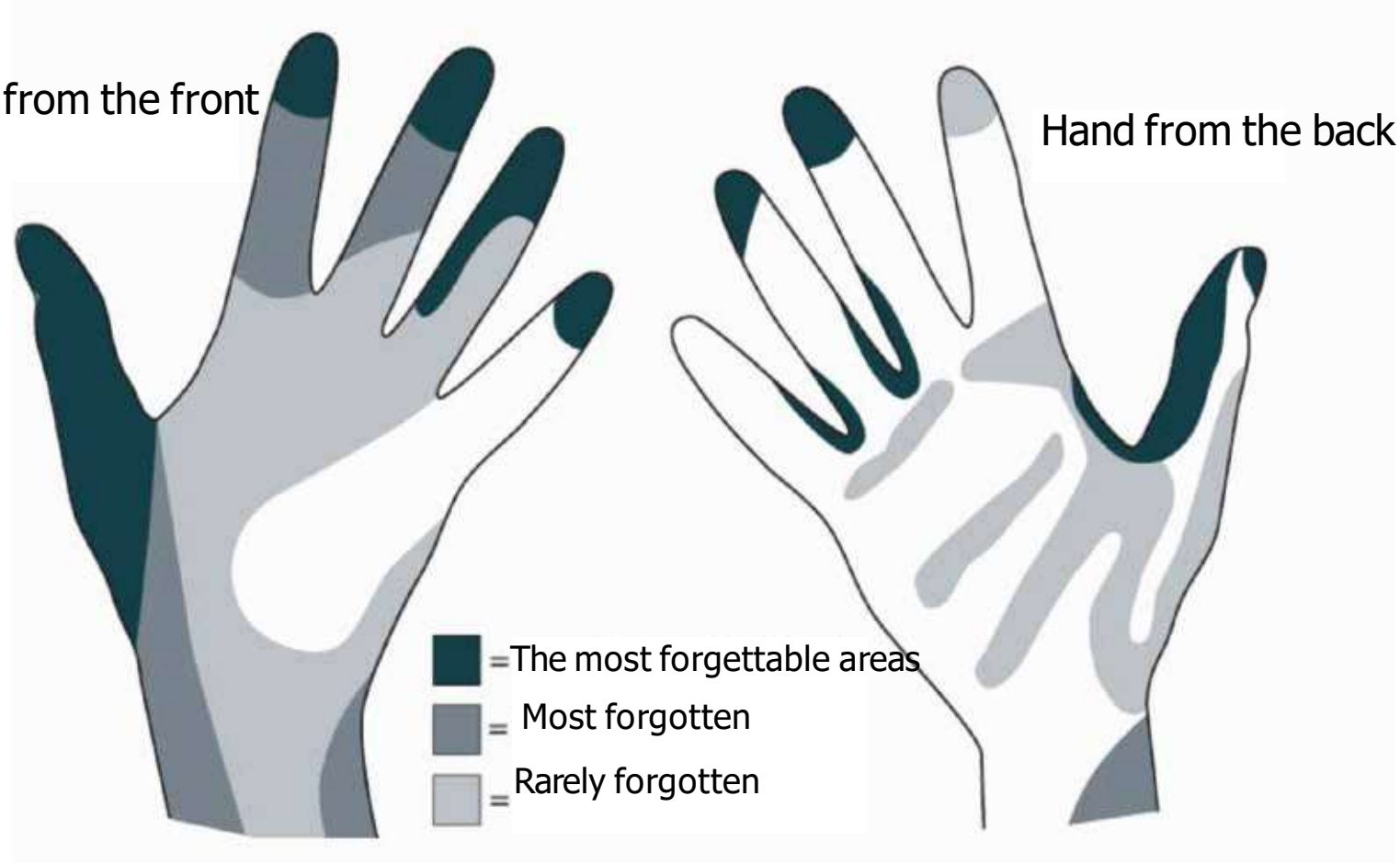


10

تجفيف الايدي

Drying hands

The most forgotten areas during hand washing



Hand wash basin characteristics

- Adequate number of hand wash basins
-Hand wash basins placed in the appropriate locations
- Must be used for hand washing only
-Wash basins must include warm water approx. 45 - 50℃
-Antiseptic soap and paper towel or drying equipment must be provided



Duties of the supervisor

- Training, mentoring, motivation
- A good role model/example
-Provide soap and cleaning equipment
- Monitoring
-Continuous evaluation of control procedures
- Take corrective actions



Requirements for protective clothing

- Comfortable.
- Light color
- No external pockets.
- No buttons so as not to fall inside the food.

-Washed inside the facility or contracted with approved laundry.

- Covers personal clothing.
- Hairnets.

-Apron not worn outside the food preparation halls.

-Provide suitable cabinets for them to keep their personal items

- Aprons are taken off when going to toilets



Symptoms that the employee must notify the supervisor of

- Diarrhea, vomiting, food poisoning
- Exposure to illness while traveling
- Eating suspicious food
- Wounds and boils
- Fever
-Infection of a family member with an infectious disease

What is the supervisor's role for ensuring effective handwashing?

- Train, instruct, motivate and effective supervision
- Lead by example/demonstration
-Systems in place for soap/towel replenishment/ running warm water
- Facilities clean and satisfactory (check)
- Monitor –visual checks/glo-term/hand swabbing
- Evaluate controls
- Corrective action, e.g. retrain/warn/dismiss



Module 6



Risks and their control from Receiving till Serving of Food

Definitions

Control measures

Procedures to prevent food safety risks.

Critical Control Points

Points in the production chain where control measures must be put to prevent/control the occurrence of hazard

Corrective actions

The actions that are taken when there is a deviation in the critical control points or control points

Monitoring

Supervisory processes to ensure that work is under control, and does not breach critical boundaries.

Rules for food safety

- Food must be protected from contamination and from pests.
- Food should not be kept at temperatures that may lead to risks to health
- Hot food to be cooled as soon as possible to avoid bacterial growth at the danger zone temperatures (can you remind us what is the danger zone?).

What are the risks and control measures of receiving food?

Risks: Contamination and multiplication

Control measures:

- Dealing with an approved supplier
- Cover and protect food during transportation and unloading
- Keep food at appropriate temperatures (chilled below °٥C, freezing below °١٨-C)

-The duration of the unloading to the final storage should not exceed ١٥minutes to avoid the food staying for a long time in an inappropriate place and temperature.

Control measures:

- Measuring the temperature of food.
- Check the hygiene of food, packaging, vehicle and driver.

Corrective actions

Refusal to receive the food in case of non- conformities to food safety.



What are the risks and control measures of the dry food storage?

Risks: Contamination and multiplication

Control measures:

- Storage in a dry, clean place.
- Proper ventilation.
- Off the floor and the wall
- Stock rotation.



Control measures:

-Monitor and verify the production and expiry date of products.

Corrective actions

Discard of expired, spoiled or unsafe food

What are the risks and control measures of refrigerated storage of food?

Risks: Contamination and multiplication

Control measures:

- Maintain the temperature at 4 °C or lower.
-Maintain stock rotation, food received first at the front, and the one received at end in the back (first in first out).
-Separate raw food from ready-to-serve food, and place raw food at the bottom of the refrigerator.
- Keep food properly covered .
- Keep the refrigerator door closed when not in use.
- Keep the refrigerator clean and avoid overcrowding.

Control measures:

- Check the temperature every four hours.
- Inspect production/expiry date and shelf life of the products

Corrective actions

Discard expired, spoiled or unsafe food



What are the risks and control measures of frozen storage of food ?

Risks: Contamination and multiplication

Control measures:

- Maintain the temperature to 18- °C or below
-Maintain stock rotation, food received first at the front, and the one received at end in the back (first in first out).
- Avoid storing food in the freezer while it is hot.
-Separate raw food from ready-to-serve foods, and place raw food at the bottom of the freezer.
- Keep food properly covered .
- Keep the freezer door closed when not in use.
- Keep the freezer in order and avoid overcrowding.

Control measures:

-Check the temperature every four hours,
check the temperature the freezer placed in dry store every 12 hours.
- Inspect production/expiry date and shelf life of the products.



Corrective actions

Disposal of expired, spoiled or unsafe stored food

What are the risks and control measures of food preparation?

Risks: Contamination and multiplication

Control measures:

- Reduce food handling to the minimum.
- Ensure adherence to the best hygiene practices.
- Use color coded utensils for raw and ready-to-eat food
- Continuous cleaning.

Control measures:

- Make sure the storage of food is at safe temperatures.
- Inspect production/expiry date and shelf life of the products.



Corrective actions

Discard expired, spoiled or food that is kept at room temperature for more than two hours.

What are the risks and control measures of cooking and re-heating?

Risks: Survival and multiplication

Control measures:

- Cook the food to reach a safe temperature more than 75 C (internal temperature).
- Take precautions to avoid contamination of food.
- Avoid reheating food more than once.

Control measures:

-Check the cooking temperature, color, shape and texture of the food and the presence of steam.



Corrective actions

Continue cooking until the food reaches the recommended core temperature.

What are the risks and control measures of hot holding?

Risks: Contamination and multiplication

Control measures:

- Ensure that the temperature of the hot holding cabinet is not lower than 63 °C.
-Take precautions to avoid contamination of food by properly covering food containers.
- Stir the food to make sure the heat is distributed evenly
- Make sure the cabinet is preheated before placing the food

Control measures:

-Check the cooking temperature, color, shape and texture of the food, and the presence of steam. the food and the presence of steam.

Corrective actions

Discard food that remained at a temperature less than 63 °C for more than two hours.



What should you do if you find frozen chicken at a temperature of 2-C?

- Inform the supervisor
-Complete the process of defrosting and cook the chicken immediately if possible or discard it.
- Ask maintenance to check the freezer



Design a list of rules for safe defrosting and cooking of 2 kg chicken with the most prominent risks in mind:

Key Risks:

- Cross contamination of other foods.
-Not eliminating microorganisms effectively as a result of not cooking or defrosting at a recommended temperature and time.
- Multiplication of microorganisms after cooking.

Defrosting/thawing requirements:

-Separate the chicken from the rest of the food during the defrosting process, if defrosting is being done inside the refrigerator place the chicken in the bottom shelf and with proper covering to avoid contamination of the rest of the food items.
- Do not wash chickens
- Clean the area after thawing.
- Make sure to cook until the internal temperature of the chicken reaches 75 °C.
- Avoid cooked chicken with raw food items.

Safe ways to defrost:

- Inside the refrigerator.
- In the microwave.



What is the importance of stock rotation?

- Make sure that the old stock is used first.
- Avoid expiry and spoilage of food and loss of stock.
- Ensure a consistent level of quality.
- Reduce the risk of the presence of pests (insects, rodents).



Shelf-life labelling



Use by:
Food should not be consumed after this date.

Best by :
: It is preferable to consume the food before the mentioned date.
Products are of lower quality after this date.

According to the laws of the **Kingdom of Saudi Arabia**, it is forbidden to display products after the expiry date or shelf-life period even if the label is preferably consume it before or best by.
The shelf life of the product depends on the extent of compliance to the conditions of transport and storage recommended to keep the product from damage and spoilage.

Module 7



Food spoilage and food preservation

Food spoilage

Control measures:

Temperature (T)

Boil for a little while then cool

(B)

Stock Rotation (SR)

Factors of food spoilage

Mold & Yeasts (T-SR)

Enzymes (T-B)

Bacteria (T-SR)

Over ripening

Physical damage

Contamination

Pests

Insects

Signs of food spoilage

Bad odors
Color
change
Viscosity
Mold
Change of texture
Deterioration of taste
Signs of pests
Stinking
Cans or packaging that are inflated or damaged
Gas Production

What is food preservation?

It is the processing of food so that the spoilage or growth of disease-causing agent is avoided or delayed.



Prevent the multiplication of bacteria inside food by:

High temperatures above 63
Low temperatures below 5C
Drying (reducing water activity)
Add salt/sugar
Chemicals (preservatives)
Addition of acid (such as adding vinegar in pickles)
Vacuum packaging
Smoking
Radiatio
n



Pasteurization, sterilization and High Temperature processing

Pasteurization:

Lower temperature and time than of sterilization

It eliminates the pathogens and some food spoilage bacteria.

Food shelf life is less, spoilage is more probable than other preservation methods.

Vitamins and nutritional value are reduced to a lesser degree than other methods of preservation.



Pasteurization, sterilization and High Temperature processing

Sterilization:

Done at a temperature above ℃100C

It eliminates all microorganisms and spores (vesicles).

Food shelf life is more than pasteurization.

Reduces vitamins and nutritional value to a higher degree.

Leads to a change in texture and taste.

Pasteurization, sterilization and High Temperature processing

High heat treatment:

It is carried out at a very high temperature for a short time.
It eliminates all species of microorganisms.
Food shelf life is the longest
Reduces vitamins and nutritional value to a lesser extent.
Do not cause a change in texture and taste.



Module 8

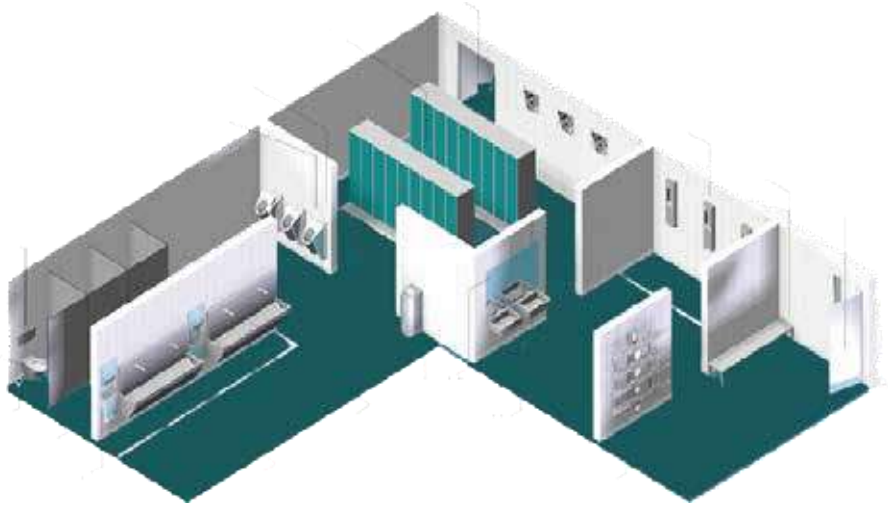


Design and use of food premises and equipment

Site Selection

Essential hygiene services:

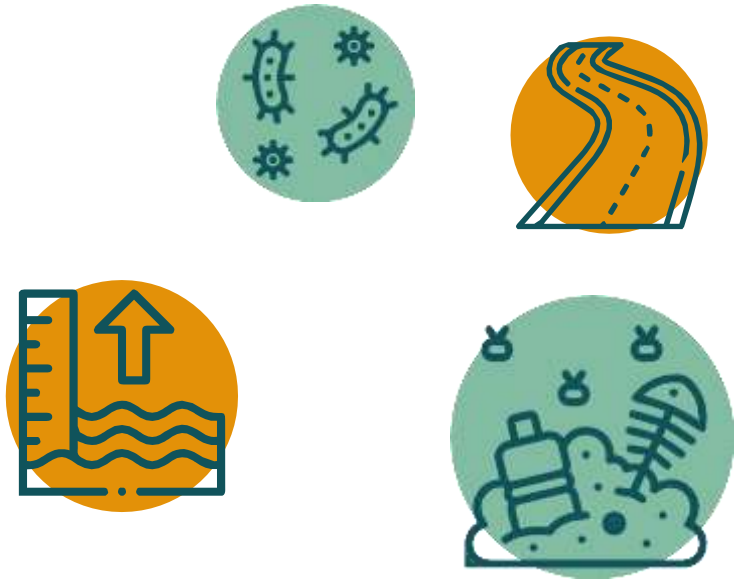
- Electricity
- Gas
- Water supply
- Effluent disposal (drainage)
- Waste disposal
- Roads
- Transport



What problems should be considered when selecting a site for a food business?

Problems to avoid :

- Flooding
- Chemicals
- Odor
- Dust
- Pests



The law relating to food premises

- Clean, in good repair and condition
- Good design, layout and construction
- Permit cleaning & disinfection
- Protect against contamination
-Permit good food hygiene practice, especially pest control
- Provide suitable temperature controlled conditions

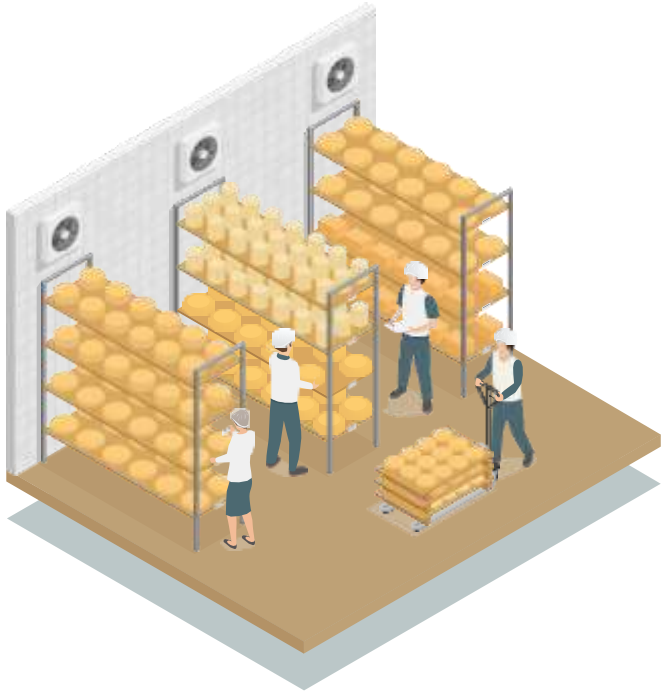
Design of food premises to reduce risk of contamination

General principles of design
- Segregation of clean and dirty processes
- Linear workflow reduces cross-contamination risk
- Personal hygiene facilities (suitable and sufficient)
- Washing and disinfection facilities (suitable and sufficient)
- Temperature control facilities
- Readily cleanable surfaces



Design of food premises

- Pest precautions
- Yards and roads well surfaced
- Staff welfare facilities
- Good lighting
-Suitable ventilation to reduce temperature and control humidity

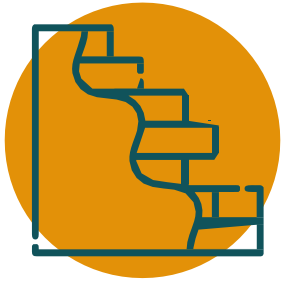


Construction of food premises

- **Ceilings**
- **Walls**
- **Ventilation:**
  - Extracts cooking fumes and grease/stream
  - Reduces temperature
  - Do not need to open windows
  - Reduces condensation
- **Lighting**
- **Doors**
- **Windows**
- **Floors**
- **Services**



Properties of surfaces



Wall surfaces

- Waterproof
- Hardwearing, non-flaking
- Easy to clean
- Smooth
- Light colored
- Cleansable joints/fixings



Floor surfaces

- Hard wearing
- Easy to clean
- Waterproof
- Jointless or cleansable joints
- Anti-slip

Why replace chipped/damaged surfaces?

- Difficult to clean/disinfect
- Risk of physical contamination
- Legal requirement
- Build -up of dirt/grease
- Harborage for insects/bacteria



What washing, waste and welfare facilities should be provided?

- Personal hygiene
- Food washing
- Cleaning and disinfection
- Cloakrooms and lockers
- Toilets
- Waste disposal (internal and external)
- Drainage

The law relating to waste

- Waste must not accumulate in food rooms
- Waste must be deposited in closable containers



Storage and disposal of waste

Internal:

- No accumulations
- Emptied frequently
- Refuse collectors
- Care when siting

Containers:

- Cleanable
- Strong
- Impervious (waterproof)
- Foot-operated lid
- Polythene sacks
- Waste disposal units
- Emptied frequently



Storage and disposal of waste

External:

- Separate from food rooms
- Pest proof/resistant
- Stored off ground
- No accumulations
- Impervious (waterproof) base
- Good drainage
- Covered

Containers:

- Cleanable/impervious
- Tight-fitting lids
- Strong/secure
- Cleaning facilities
- Emptied regularly
- Suitable size



What are the characteristics of food equipment?

- Clean, in good repair and condition
- Minimize the risk of contamination
- Enable thorough cleaning and where necessary, disinfection
- Be installed to allow cleaning of the surrounding area



Module 9



Cleaning and disinfection

Cleaning is the systematic application of energy to a surface or substance with the intention of removing dirt

What risks come from cleaning?

- Cross- contamination
- Microbiological hazards.
- Chemical hazards
- Physical hazards.
- Allergenic hazards
-Failure to remove/destroy microorganisms



Definitions

Bactericide

A substance that kills bacteria

Biodegradable

A substance that can be decomposed/degraded by bacteria or other organisms

Contact time

Time of chemical application on the surface to ensure its effectiveness

Definitions

Detergent

A chemical used to remove dirt and grease and food particles but does not destroy bacteria

Pesticide

A chemical used to reduce living organisms to a safe level

Sanitizer

Combined detergent and disinfectant

Sterilizing

The process of destroying all microorganisms and toxins and spores

Sterilizing chemical

A chemical used to destroy all objects .micro-organism, germs, toxins

What are the benefits of disinfection and cleaning?

- Reduces the risk of food poisoning
- Remove bacteria from food items
- Removes substances that attract of pests
- Removes foreign substances that may lead to contamination
- Removes dirt and grease
- Reduces customer complaints
- Provides a safe environment for employees and customers
- Allows disinfecting
- Complies with the law
- Promotes a favorable image

Energies in cleaning

Kinetic

- Physical- scrubbing / brushing
- Mechanical –machines
- Turbulence - running water



Thermal

- Hot water



Chemical

Detergents –remove grease and other soiling
Available as:
- Powder
- Liquids
- Gels
- Foams



Disinfection

The reduction of micro-organisms to a safe level Can be achieved by using:

- **Hot water**
- **82 C for 30 seconds**
- **Steam**
- **Dishwasher**



What requires cleaning and disinfection?

- Hand – contact surfaces
- Food – contact surfaces
- Cleaning materials and equipment

What prevents effective disinfection?

- Presence of food scarps, soil and detergent
- Inadequate contact time
- Wrong concentration

Forms of cleaning agents:

Foam	gel	liquid	powder

How do we clean?

- In a superficial way:

Reduce surface tension to enable the detergent to penetrate dirt.

- By dispersion:

It elevates/removes dirt from the surface.

What should be both cleaned and disinfected?

Tool	Cleaning and disinfection
Knives and forks	✓
Inside the oven	
Inside the refrigerator	✓
Kitchen floor	
Cutting board	✓
Ceiling	
Hand wash basin	✓
Drawer handle	✓
Refrigerator door handle	✓

Tool	Cleaning and disinfection
Pans	
Shelves used for packaged dry goods	
Shelves of display refrigerators	✓
Potato peeler	✓
Cheese grater	✓
Hand broom	
Fabric detergents	✓
Protective clothing	✓

How can you tell if surfaces have been disinfected effectively?



It is determined by bacteriological examination (Swab) and ATP. **Not visual inspection.**

Why to clean and disinfect surfaces, in contact with food as soon as possible, after use?

To prevent the accumulation of food residues and prevent the growth and multiplication of bacteria .

Six steps to cleaning

01



Dirt and particles removal

02



Main cleaning

03



Rinsing

04



Disinfection

05



Final rinse

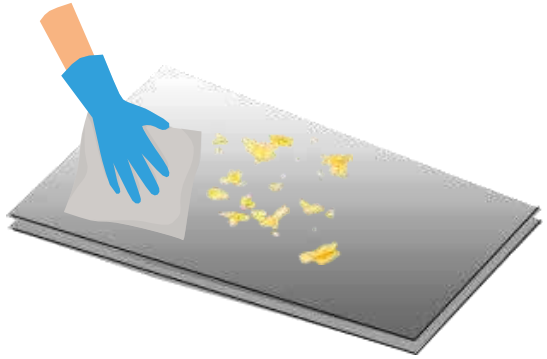
06



Drying and storing to prevent contamination.

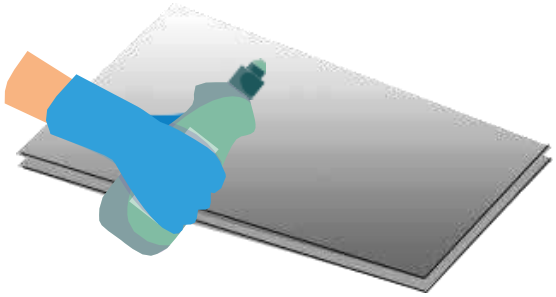
Stages of sanitizing

01



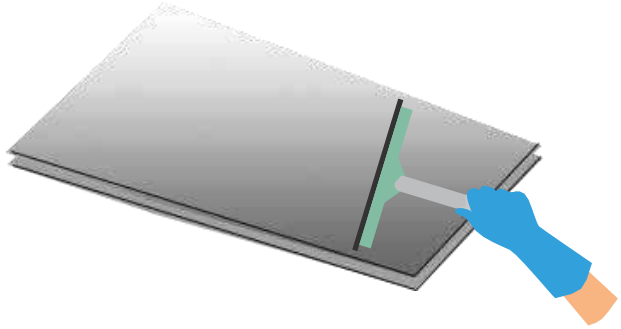
Dirt removal and cleaning

02



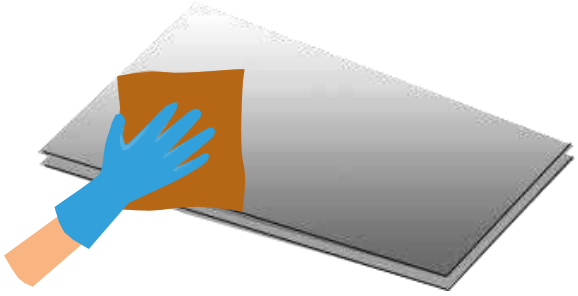
Sanitizing

03



Rinsing (optional)

04



Air drying

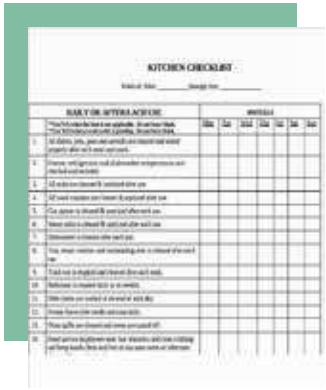
cleaning schedule

- Essential to ensure effective cleaning and disinfection

Surface

- What
- How
- When
- Time allowed

- Cleaning schedules must be clearly written


KITCHEN CHECKLIST

The cleaning schedule must also contain



-The name of the person who cleaned up.
- Protective clothing used.
- Safety procedures followed.
- Checklist.

The cleaning schedule should contain the following:

- Chemicals used
- Type
-Concentration
- Storage

Cleaning checklist should be clearly written and verified and signed after completion.



How can a supervisor ensure cleaning procedures are satisfactory?

- Providing clear instruction
- Suitable training (to ensure competent staff)
- Provide/check cleaning schedule
- Suitable materials
- Equipment and time
- Motivate
- Observing and monitoring staff undertaking the cleaning activities



How can a supervisor monitor standards of cleanliness? Daily visual checks/white paper tissues/ATP

How can a supervisor verify effective disinfection (safe cleaning)? Microbiological swabbing

Module 10



Pest Control

What is a food pest?

An animal, insect or bird which lives in or on our food. It contaminates food and is noxious, destructive or troublesome.

How do pests contaminate food?

- Breading in decaying matter
- Feeding (vomit back previous meal)
- Feces
- Walking on it/work surfaces
- Laying eggs on uncovered food
- Dead bodies

Food Pests

Rodents

- Brown rat
- Black rat
- House mouse

Insects

- Flies
- Wasps
- Stored product insects

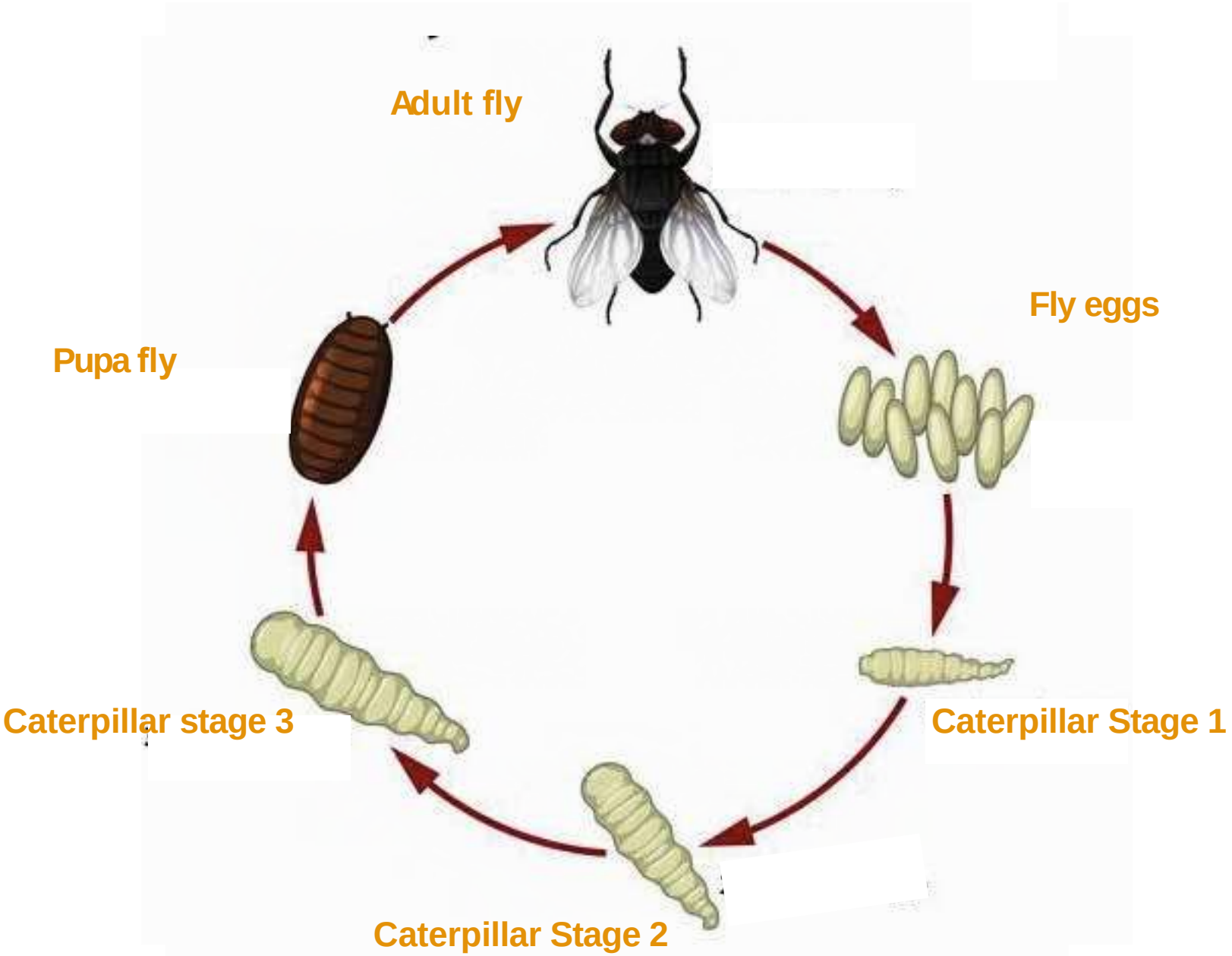
Birds

Dogs

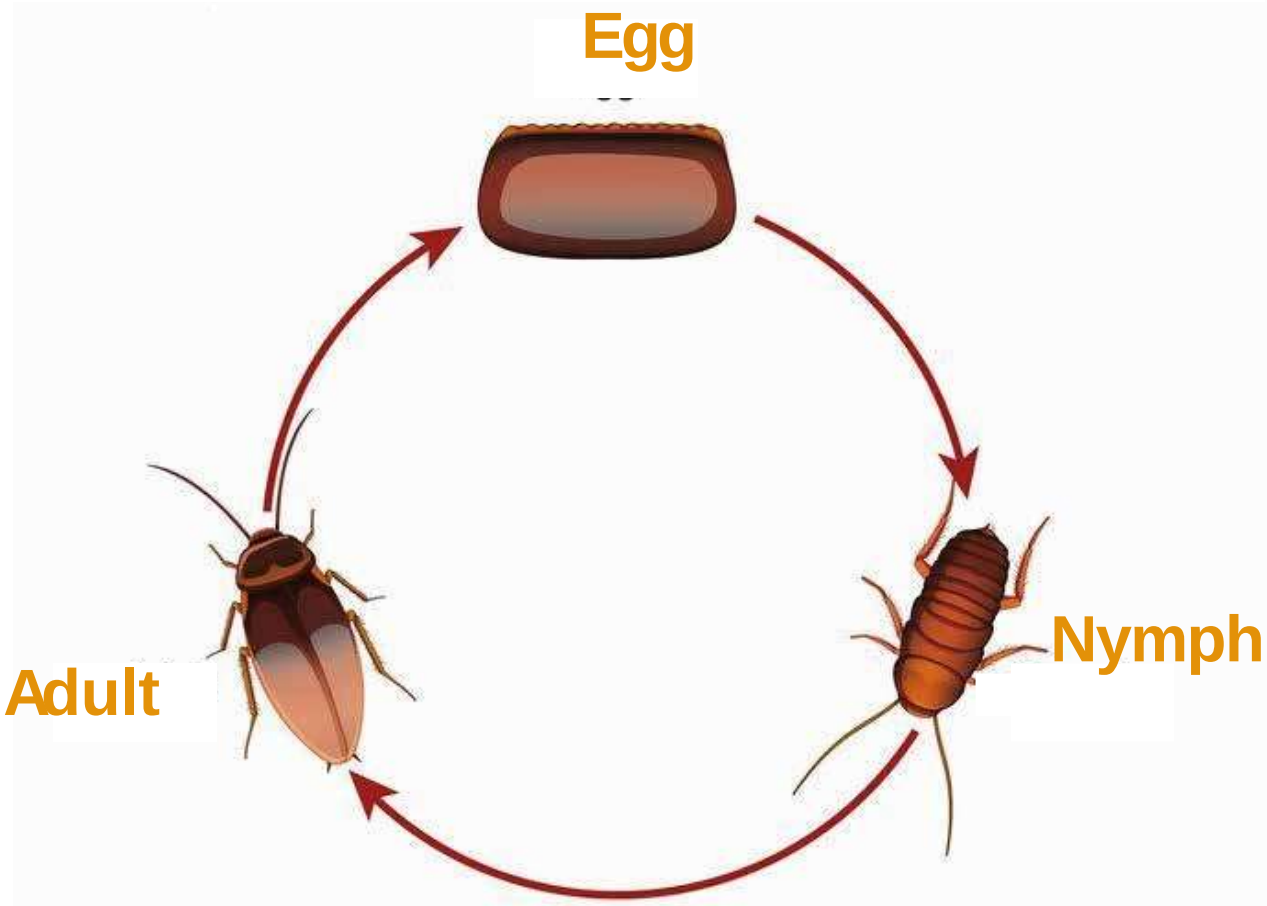
Cats



Life cycle of the fly



Life cycle of cockroaches



Life cycle of rodent

Adults



Adolescene

Mother and babies

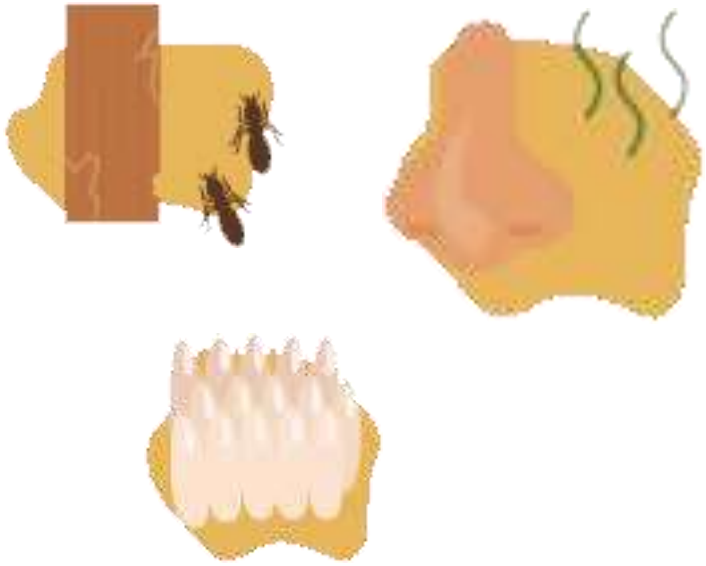
What are the signs of rodent infestation?

- Bodies
- Droppings
- Smears
- Footprints/tail marks
- Runs
- Holes
- Chewed packaging
- Chewed food/paper
- Gnawing damage
- Nests
- Bait takes
- Noise
- Hair/fur



What are signs of insect infestation?

- Bodies/nymphs
- Larvae/pupae
- Eggs
- Egg cases
- Smell
- Frass (dust piles)
- Webbing
- Noise
- Holes in food e.g. biscuits.



What ate the reasons for pest control?

- Buildings must be kept clean and especially waste areas
- Do not leave any food outside
- Any food residues must be removed immediately
- Store food in pest-resistant containers
- These are all part of good housekeeping

- Cut back vegetation
- Remove unwanted items (harborage)

How can you avoid attracting pests?

- Buildings must be kept clean and especially waste areas
- Do not leave any food outside
- Any food residues must be removed immediately
- Store food in pest-resistant containers
- These are all part of good housekeeping

- Cut back vegetation
- Remove unwanted items (harborage)



Environmental Control To Prevent Infestation

Good Housekeeping (entry)

- Protection of food
- Cleanliness
- Remove spillages
- Remove harborage
- Manage waste
- Staff training

Prevent access

- Maintenance of drains/buildings
- Keep doors closed (A&I)
- Keep windows closed (I)
- Air Curtains (I)
- No gaps under external doors (A&I)

A- Animals I- Insects

Eradication

Physical control

- U/V fly killers –best method
- Glue board recommended
- Siting critical
- Cockroach monitoring traps
- Rodent traps (live or dead)
- Hormone traps
- Sticky flypapers

Preferred to chemical control as PEST CAUGHT DEAD or ALIVE

Eradication

Chemical control

- Rodenticides –solid blocks, paste, bait and powders
- Insecticides (acute or chronic) –knockdown and residual, dust, baits, gels and sprays.
- Fumigation
- Narcotising

Risk of contamination

- Dead pests (inaccessible places)
- Pesticides

Use reputable contractor

What is the role of a supervisor in pest management?

- Proactive pest management
- Inspect vulnerable areas
- Signs (advise and instruct staff)
- Ensure staff report signs of pests
- Ensure deliveries checked
- Notify contractor if evidence of infestation
- Ensure contractor does not contaminate food

•Ensure necessary cleaning undertaken following contractor's visit

•Ensure necessary cleaning undertaken following pest activity

What is the role of a supervisor in pest management

- Report defects
- Accompany contractor
- Maintain high standards of hygiene
- Follow contractor's recommendations
- Record bait box positions
- Check bait boxes regularly
- Ensure contractor's visiting records are maintained

How can you evaluate the effectiveness of pest control?

•Absence of pests/signs of pests- pest control book (use traps to monitor)

- Glue board fly killers empty
- Evidence of proofing/maintenance
- Good housekeeping
- Visual check –food storage
- No complaints –staff public

What should a supervisor do if he finds evidence of mice in a food room?

- Stop any food being sold and isolate any contaminated food so there is no risk to other food (may need to immediately dispose of food, although evidence may be required for contractor)
  - Call contractor
  - Ensure area is thoroughly cleaned and disinfected
- Act on recommendations of contractor, especially regarding proofing

Module 11



HACCP – Hazard Analysis Critical Control and Food Safety Management Systems

What is the hazard analysis critical control point system?

- Preventive system that deals with food safety.
-Determining the risks to food safety whether biological chemical or physical.
-Identifying critical points CCPs that need to be controlled to ensure product safety.

What are the initial requirements for the implementation of a HACCP system?

Prerequisite programs are good hygiene practices the company must have before implementing a HACCP system – Hazard Analysis Critical Control Point

1. Good design
2. Preventive Maintenance
3. Personal hygiene
4. Glass Policy
5. Stock rotation
6. Cleaning and disinfection
7. Pest Management
8. Approved Supplier
9. Allergen policy
10. Waste Management
11. Traceability
12. Emergency Plan
13. Training and Supervision



The Seven Principles of the HACCP System

Principle I: Conduct a hazard analysis

•Identify the hazards that are likely to occur in proportion to a reasonable probability if not controlled.

•Determine the production steps that food goes through from start to finish.

Principle II: Identification of Critical Control Points

•Specify the production steps at which it is possible to control the risks identified in the first principle, by eliminating them or by reducing them to the acceptable level.

Principle III: Establishment of Critical Limits

Critical limits are the boundary between the acceptable and the unacceptable.

The Seven Principles of the HACCP System

Principle IV: Development of methods of monitoring

•Monitor critical control points and critical limits to make sure that these points are under control and within the safe limits.

Principle V: Development of corrective actions

•Done when one of the critical control points goes beyond the “safe range"

• Requires a prior plan to counter the loss of control

• Designed to minimize damage

•Might be used for continuous improvement by documenting each problem that were not previously mentioned in the pre-established plans

The Seven Principles of the HACCP System

Principle VI: Development of a verification system

This rule aims to ensure the validity of the system

Principle VII: Development of a documentation system

- The HACCP system requires documentation in written form or in any other way.
- Records should be simple and easy to manage

What are the benefits of HACCP?

- Prevention of diseases transmitted by Food
-Enhances the brand/image of the company
- Compliance with the law
- Low costs
-Reveals your commitment to the production/sale of food
-Increases consumer/customer confidence and merchant, retail & government Bodies



What is a flow chart and hazard analysis?

Flow Chart

It is a pictorial representation of the steps involved in a particular process.

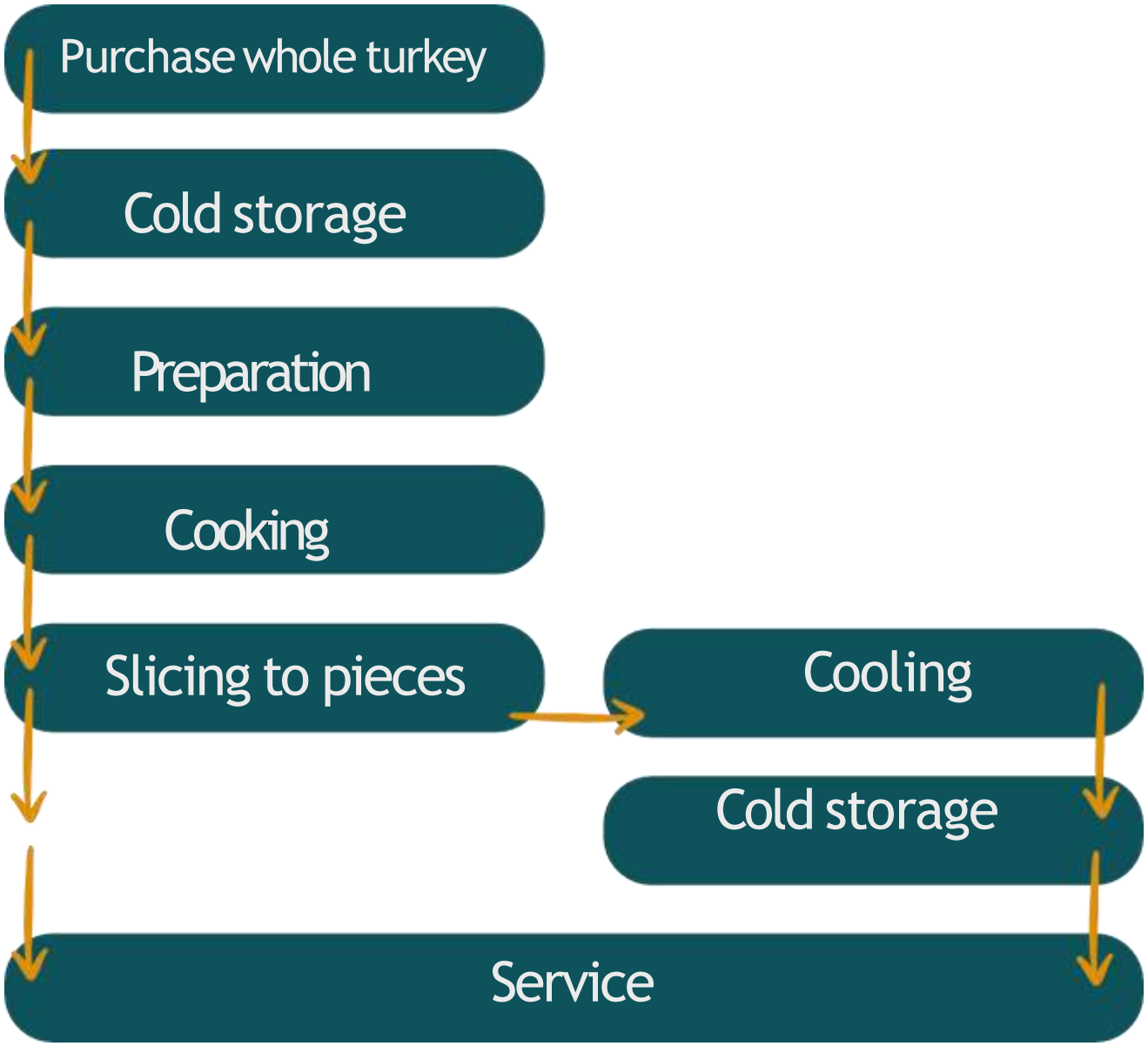
What is hazard analysis?

•Evaluating information on hazards and conditions leading to their presence to decide those which are significant to food safety.
•Analysis of a food process to determine where, if things go wrong, the food products made may be harmful.

Consider:

- Contamination, multiplication, and survival (toxins/spores) at each step
- Raw ingredients
- Characteristics/ composition
- Processing/preparation
- Microbiology
- Premises/equipment design
- Packaging
- Distribution
- Intended use/consumer

Product flow chart



Identify the hazards (Principle 1)

- A hazard is anything with the potential to cause harm
- Risk is the likelihood of the hazard occurring and its severity

(Micro) biological (CMS)	Physical (C)	Chemical (C)	Allergenic (C)
Food borne illness e.g. Salmonella	Cuts to mouth, chocking, broken teeth, internal injury, burning e.g. glass, nails/bolts, string, jewellery	Food poisoning, chronic illness e.g. cleaning chemicals, pesticides, weedkillers, additives, poisonous foods.	Immune reaction, anaphylactic shock e.g. peanuts, milk, eggs, shellfish, gluten, soy, sesame seeds.

Key: C= contamination, M=multiplication, S= survival

What are control measures (Principle 1)?

Action required to prevent or eliminate a food safety hazard or reduce it to an acceptable level.
Controls can be applied to:

- Temperature
- Time
- pH
- Size/shape/print
- Time
- aW
- Additives
- Appearance/texture/colour

What are the hazards being controlled by these generic controls?

- Approved suppliers (prerequisite) →
- Staff vigilance and training (prerequisite) → Contamination, multiplication and survival
- Cleaning and disinfection (prerequisite) → Contamination and survival
- Stock rotation (prerequisite) → Multiplication
- Color coding (prerequisite) → Contamination
- Effective maintenance (prerequisite) → Contamination, multiplication and survival

What are the hazards being controlled by these generic controls?

- Segregation of raw and ready-to-eat foods (prerequisite) → Contamination
- Salt/sugar/acid / adjust pH/keep dry → Multiplication
- Size, weight and shape of joint/time management/storage temperature → Multiplication
- Cooking/processing → Survival

Risk assessment matrix

Risk \ Severity	High	Med	Low
High	High Risk (3) Low Severity (1) RxS = 3 Bone in fish	High Risk (3) Med Severity (2) RxS = 6 CCP Salmonella in under cooked chicken	High Risk (3) High Severity (3) RxS= 9 CCP E.coli O157 in under cooked beef burger
Med	Med Risk (2) Low Severity (1) RxS = 2 Stone in Jam	Med Risk (2) Med. Severity (2) RxS= 4 Mouse in product	Med Risk (2) High Severity (3) RxS= 6 CCP Botulism in low-acid food
Low	Low Risk (1) Low Severity (1) RxS= 1 Cardboard in cooked chicken meal	Low Risk (1) Med. Severity (2) RxS= 2 Pin in cooked chicken	Low Risk (1) High Severity (3) RxS= 3 Razor blade in load of bread

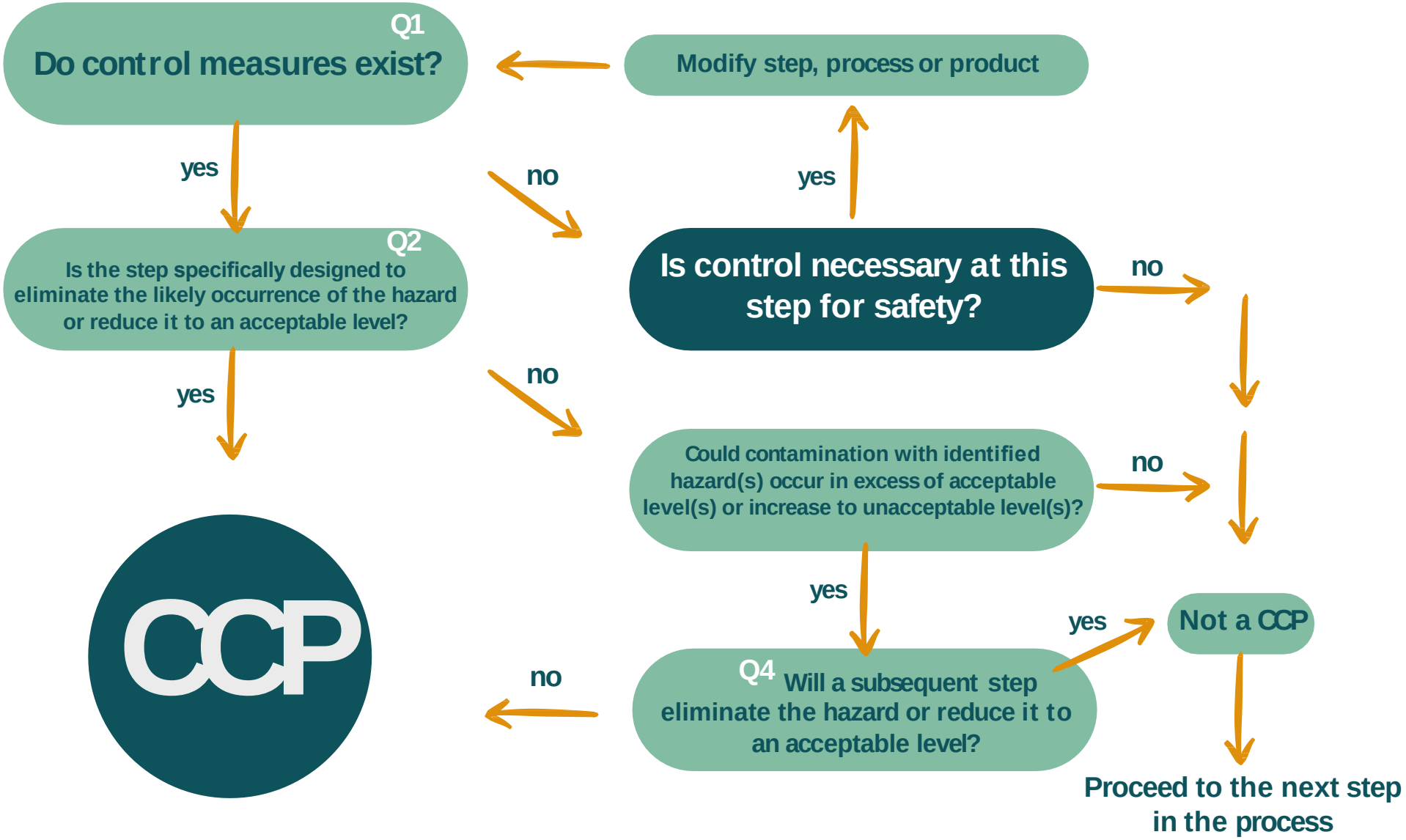
*Used to access significant hazards or to ensure critical control points are critical.

What is a critical control point (Principle 2)?

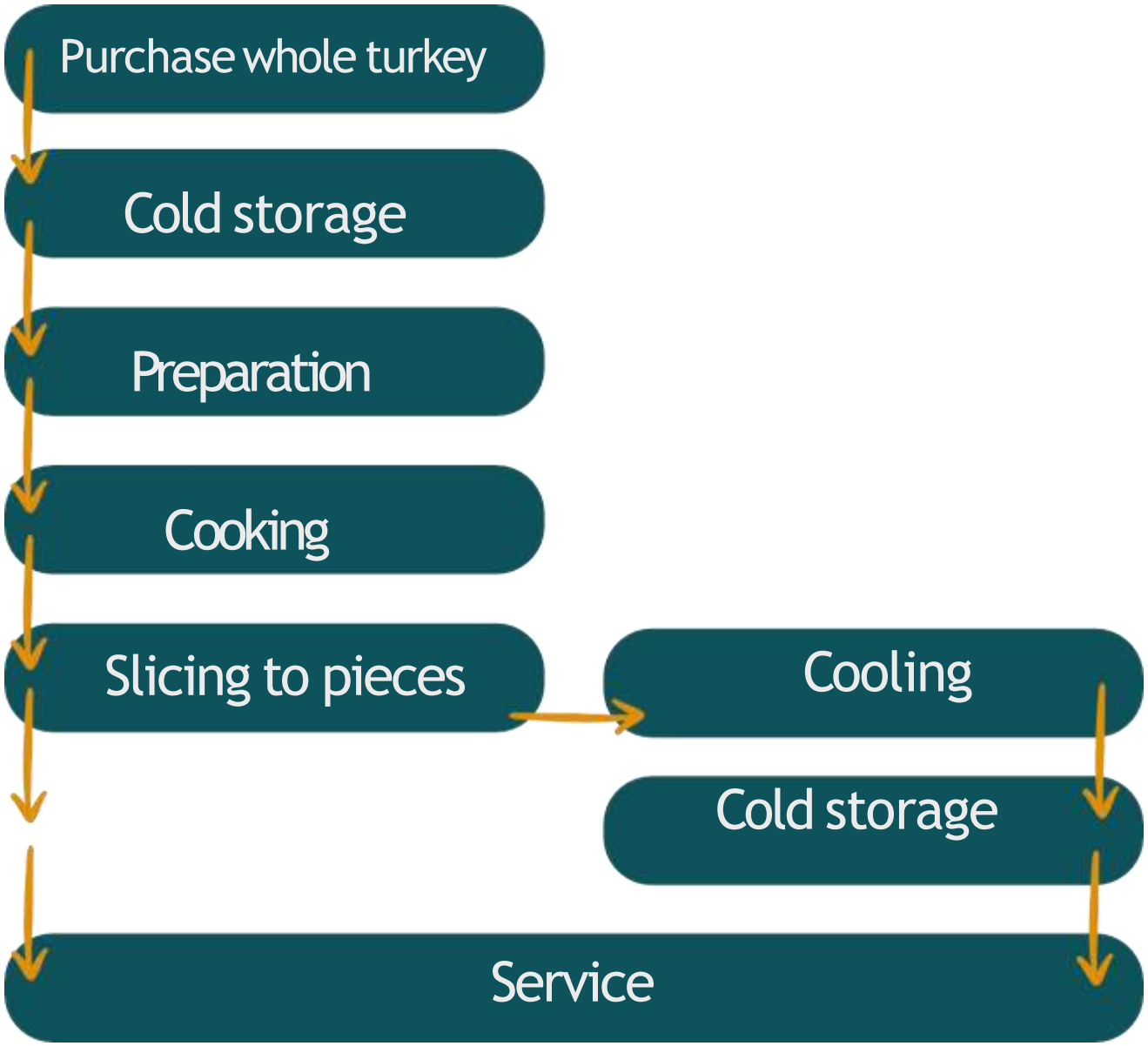
Critical control point: A step in a process where control is essential to prevent or eliminate a food safety hazard, or reduce it to an acceptable level.

Effective control procedures must be provided at all CCPs

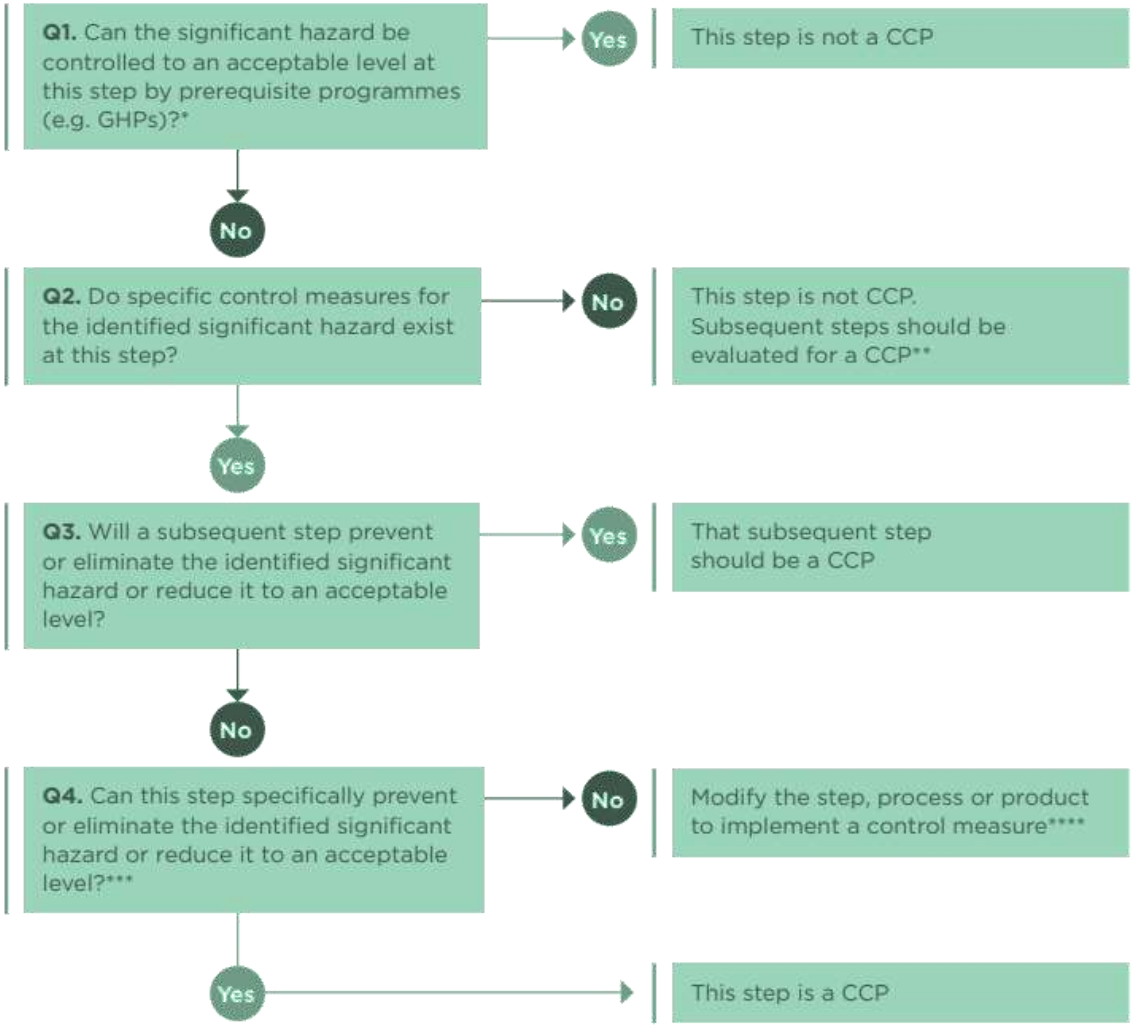
HACCP Decision Tree (Old version)



Product flow chart



HACCP Decision Tree (new version)



* Consider the significance of the hazard (i.e. the likelihood of occurrence in the absence of control and the severity of impact of the hazard) and whether it could be sufficiently controlled by prerequisite programmes such as GHPs. GHPs could be routine GHPs or GHPs that require greater attention to control the hazard (e.g. monitoring and recording).

** If a CCP is not identified at questions 2–4, the process or product should be modified to implement a control measure and a new hazard analysis should be conducted.

*** Consider whether the control measure at this step works in combination with a control measure at another step to control the same hazard, in which case both steps should be considered as CCPs.

**** Return to the beginning of the decision tree after a new hazard analysis.

What are critical limits and target levels?

C 7 → Acceptable
C 8 → Critical limit
C 9 → Unacceptable

	Critical limit	Target
Refrigerator	8 C	5 C
Cooking temperature	75 C	78 C

- Critical limits must be unambiguous and measurable
- Critical limits "values of monitored actions which separate the acceptable from unacceptable"
- Target levels "control criteria that are more stringent that the critical limits"

What is the monitoring of controls (principle 4)?

Monitoring “the planned observations and measurements of control parameters to confirm the process is under control and critical limits are not exceeded”

- Rapid detection and correction
- Automatic or manual

Types of monitoring



Measuring (temperature/weight/volume) e.g. refrigerator



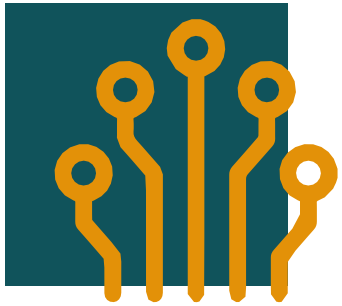
Observation/ supervision



Visual inspections (of premises, vehicles or pesticides)



Competency Testing



Organoleptic (senses)



Daily checking of controls/records (e.g. date codes)

Checking and recording food temperatures

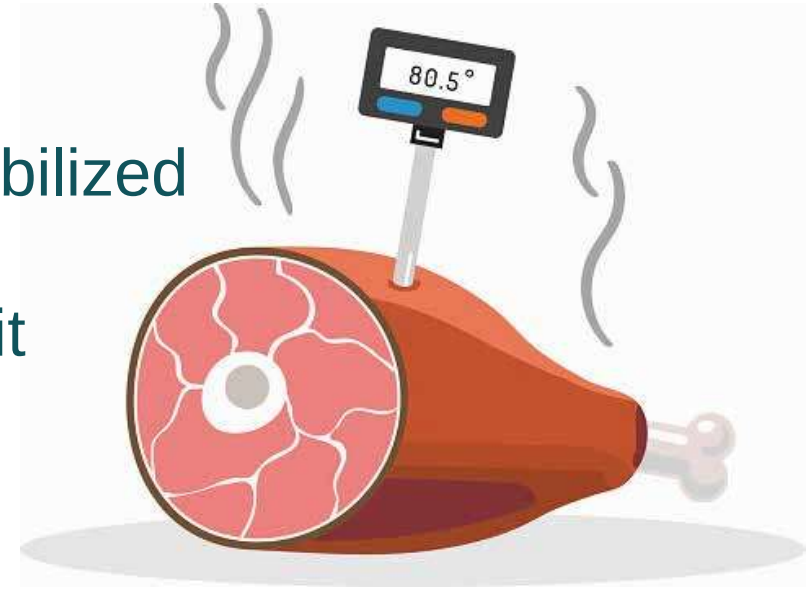
- Use a clean and disinfected probe thermometer.
- Check the temperature at the core of the food.
- Keep probe thermometer for at least 30 seconds.
- Avoid fat, bones and container extremities.
- Record temperature.



only infrared digital thermometers should be used to measure surfaces (dry or cold storage) No risk of cross-contamination. Fast but not precise.

Errors that can be made when checking temperature with a probe thermometer

- Not calibrated
- Not measuring the core temperature
- Not properly cleaned and disinfected
-Used to measure the temperature of both raw and ready to eat food, without being disinfected in between
- Touching bones or containers
-Not allowing it enough time to display a stabilized temperature
-Record a temperature with a fahrenheit unit as celcius unit



What is corrective action (principle 5)?

Corrective action: The action to be taken when a critical limit is breached. Remedial action should be taken before a critical limit is breached.
Corrective action should bring the CCP under control and deal with any affected product.



Product Outside Critical Limit

•Corrective action should specify the treatment of affected product (quarantine, testing, reprocessing, disposal and recall)

- Continue process, e.g. extend cooking time
- Change shelf life, e.g. use immediately
- Release after examination/sampling/testing
- Use for different purpose
- Release
- Destroy



What is validation and what is verification?

•**Validation:** Obtaining evidence that elements of the HACCP plan are effective, especially the critical control points and critical limits.
Verification: The methods, procedures, tests, and other evaluations, in addition to monitoring, to establish if the HACCP systems is functioning as planned.

- **Audits**
- **Analysis of complaints**
- **Microbiological/chemical tests**



Probe thermometer calibration

Cold



Ice cubes
-1 to +1

Hot



Boiling water:
99 c to 101

What should be included in opening and closing checks?

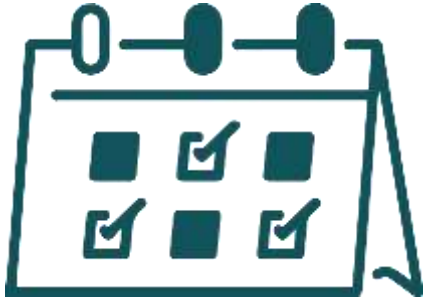
Opening checks:

- Fridges and equipment
- Staff fitness/clothing
- Hand washing facilities
- Cleanliness of surfaces
- Adequate cleaning materials
- Absence of pests/hazards

Closing checks:

- No food left out
- Out-of-date food discarded
- Dirty cloths replaced
- All waste removed
- Windows closed
- Records completed

*Demonstrates commitment to food safety



What is review?

A reassessment of the HACCPsystem to ensure it's continued validity.

The HACCP system should be reviewed:

- If things go wrong, e.g. food poisoning
- If there are significant changes, e.g.new ingredients, law, process or product.



Documentation

- Essential to the application of the HACCP system
- Appropriate to the size and nature of the business
- Demonstrates importance of CCP monitoring staff

Why is it required?

- Company policy
- Verification/internal audits
- Compliant/illness investigation
- Due-diligence defense
- Legal compliance
- External auditors/EHO/EHP
- Prove business is well managed

HACCP Records

Records include:

- CCP monitoring activities

Deviations, corrective actions and recalls

- Modifications to the HACCP system
- Audit reports
- Customer complaints/investigation results
- Calibration of instruments
- Prerequisite program: records



The law relating to food

- Food must be protected from contamination and from pests
- Food must not be kept at temperatures that might result a risk to health (adequate refrigeration is required for perishable and high – risk food)
- Hot food to be cooled must be cooled as quickly as possible

Module 12



Quality and Food Safety Policies

What is the purpose of developing a food safety policy?

-To prove that the establishment is committed to the production of safe food.
- To ensure compliance with the regulatory laws.
-A guarantee to customers by providing them with high quality products, safe and free of chemical, physical and biological contaminants.

What should present in a food safety policy?

- Management organization and responsibilities.
-Commitment to the production of safe food and the implementation of high standards.
-Commitment to comply with the law and keep abreast of the latest developments.
-Commitment to the implementation of the HACCP Food Safety Management System.
-Provision of policies related to prerequisite programs, including pest control, cleaning, and waste management
- Policy on personal hygiene and training.
-Policies on traceability and customer complaints (including food poisoning) .

Who are the persons involved in the investigation in case of food poisoning?

-Quality Control Officer
-Consultant for the control of infectious diseases



Module 13



Food Safety Legislation and Standards

- Compliance with the legal requirements and legislations of the regulatory authorities regarding the food safety system in restaurants, hospitals, food factories and food service providers.

The Saudi Food and Drug Authority contributes to food safety in Kingdom of Saudi Arabia through the preparation of integrated legislation from technical regulations to food and specifications products. It also provides courses and workshops to raise and spread awareness on the concept of food safety.



Fines and reprecautions

In case of food poisoning, the facility must bear a fine of 10,000 Saudi riyal for each affected person and the facility will beshut down.



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Thank you!

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